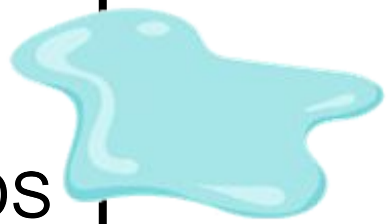
CHAPTER 5:
STATES OF MATTER

5.0 STATES OF MATTER

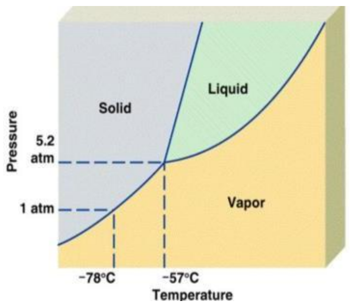
5.1 GAS

5.2 LIQUIDS



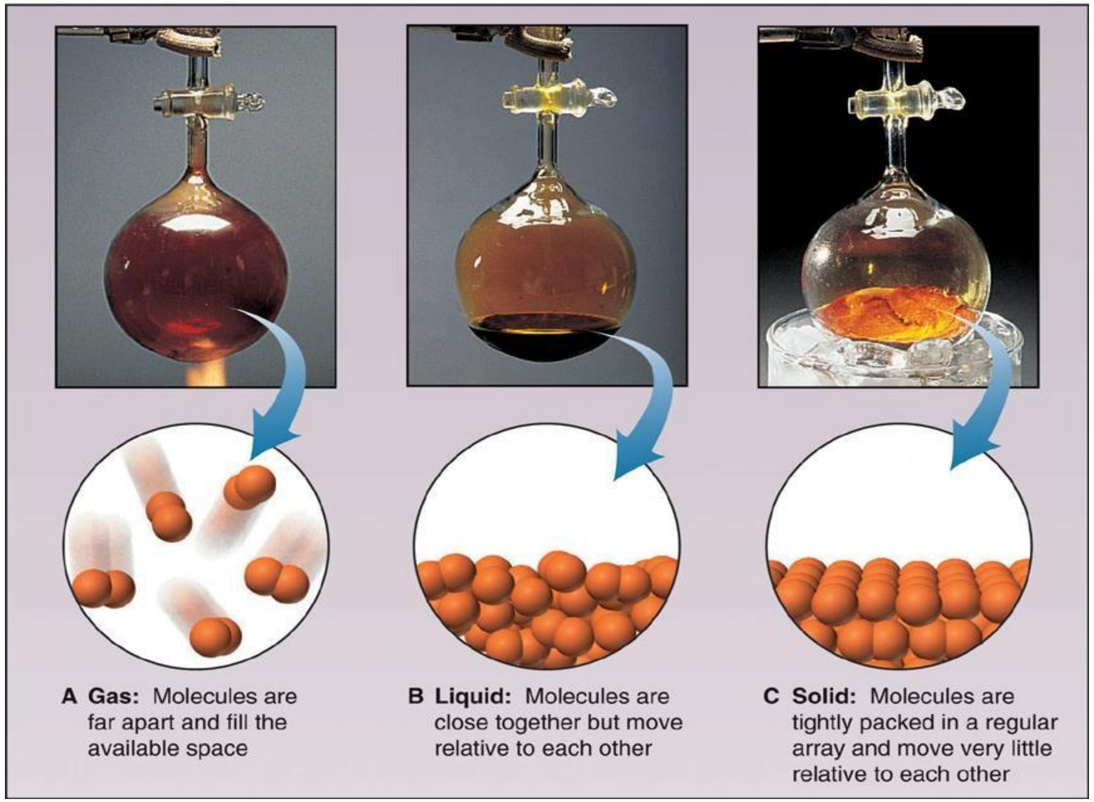
5.3 SOLIDS

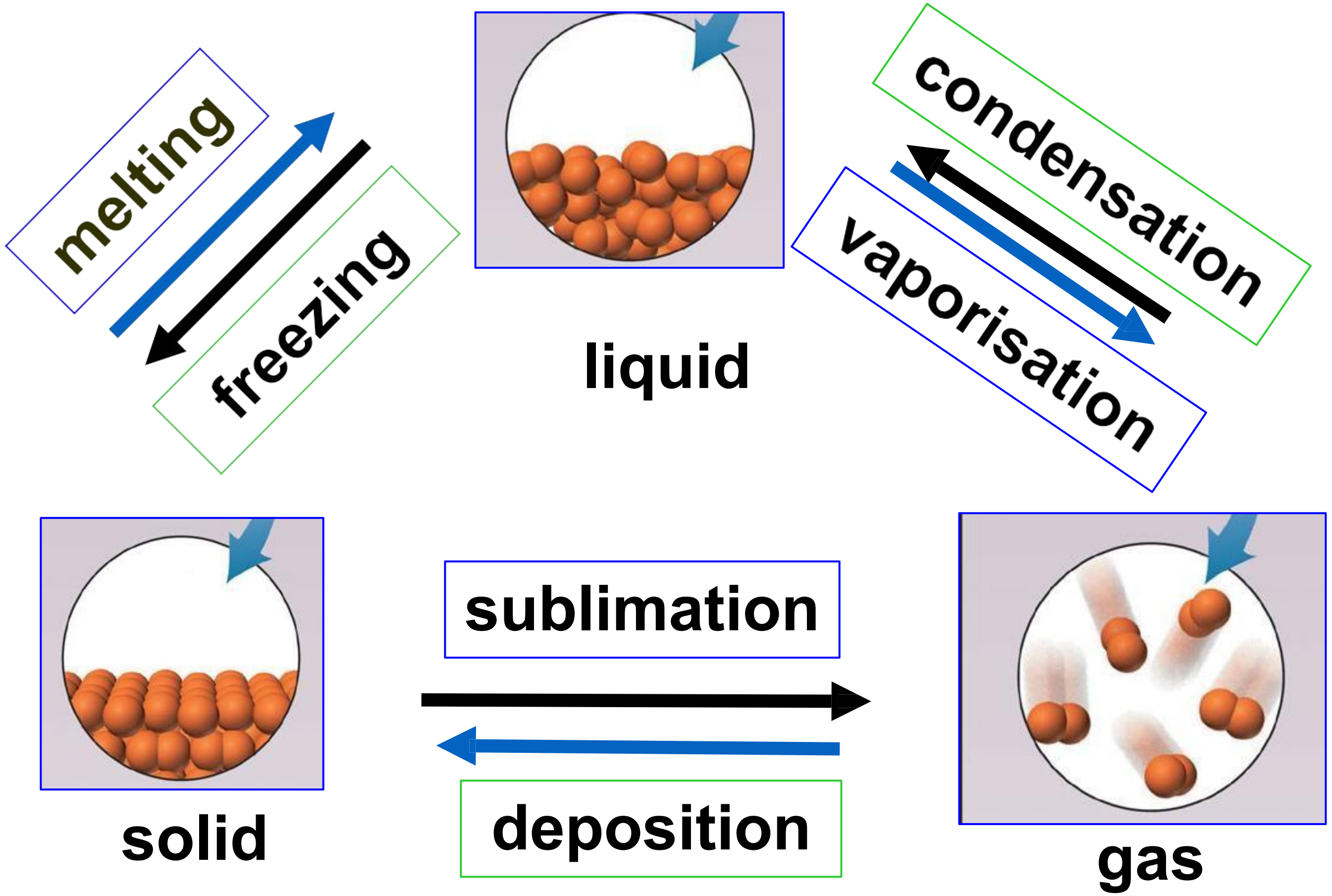
5.4 PHASE DIAGRAM



3 States of Matter :

Gas Liquid solid





Inter-convertible states of matter

- ✔ Changes in state are described as physical changes. When a substance undergoes a change in state, many of its physical properties change.

- ✔ The table summarizes the important differences in physical properties among gas, liquid, and solid.

	Solid	Liquid	Gas
Volume	Fixed	Fixed	Volume of the container
Shape	Fixed	Take the shape of the container, but not necessarily occupy the whole container	Takes the shape of the container which it occupies
Compressibility	Incompressible	Very low	High
Relative density	Very high	High	Low

5.1 GAS

✔ General properties of gas:

	Gas
Volume	No fixed volume. Takes the volume of the container
Shape	Takes the shape of the container which it occupies
Compressibility	High
Relative density	Low

✔ An ideal gas is a simplistic model that describes how molecules or atoms behave at the microscopic level.

✔ It is used to determine the relationship of temperature, volume, pressure, and quantity in a gaseous system.

✔ **An ideal gas behaves precisely according to the Kinetic Molecular Theory**

Kinetic Molecular Theory of Gases

1. Gases consist of large numbers of molecules that are in **continuous random motion**.
2. The **volume** of all the **gas molecules** is **negligible** compared to the total volume in which the gas is contained.
3. **Attractive and repulsive forces** between gas molecules are **negligible**.
4. **The collisions** between the gas molecules are perfectly **elastic**.
5. **The average kinetic energy gas molecules is proportional to absolute temperature**.

GAS LAWS

Four Variables of Gases

- **nn Moles**
- **V n Volume**
- **P n Pressure**
- **T n Temperature**

GAS LAWS

- Boyle's Law
- Charles' Law
- Avogadro's Law

Ideal Gas Law

Each gas law describes the relationship between two variables, given that the remaining two variables are held constant

Constant variable :	n & T

Boyle's Law

by Robert Boyle in 1662

- State that for a fixed amount of gas at constant temperature, gas volume is inversely proportional to gas pressure.

$$V \alpha \frac{1}{P} \quad \text{at constant n, T}$$

$$PV = k \quad \text{where k is a constant}$$

- For a given sample of gas at different pressure and volume:

$$\boxed{P_1V_1 = P_2V_2}$$

P_1 = initial pressure

V_1 = initial volume

P_2 = final pressure

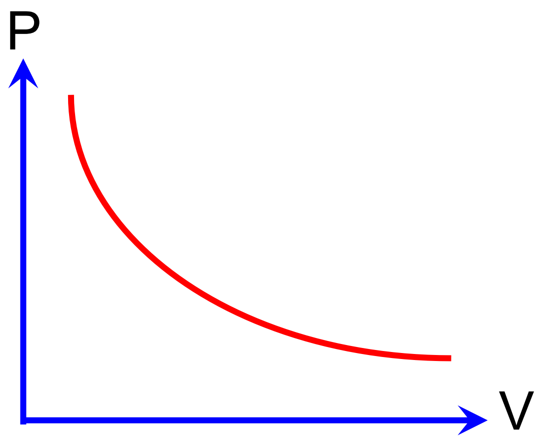
V_2 = final volume

- In practice, real gases do not obey Boyle's law because under high pressure, gases do not behave ideally and would liquefies because its volume is reduced to zero.

Constant variable :	n & T

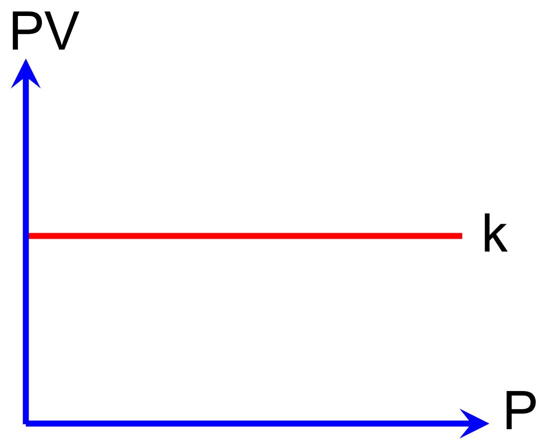
Boyle's Law can be represented by the following graphs at fixed amount of gas and constant temperature:

(a) Pressure against volume



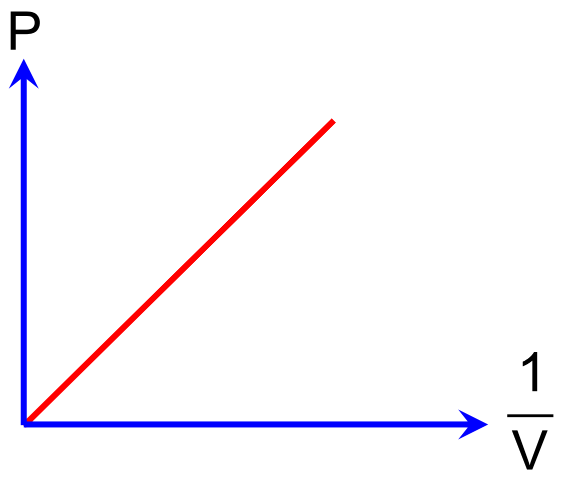
Illustrate that pressure is inversely proportional to volume.

(b) PV against pressure



Illustrate that PV = constant

(c) Pressure against $\frac{1}{V}$



Illustrate that $P \alpha \frac{1}{V}$

Constant variable :	n & P

Charles' Law

by Jacques Charles in 1787

✔ State that for a fixed amount of gas at constant pressure, gas volume is directly proportional to temperature (in Kelvin).

$V \alpha T$ at constant n, P

$\frac{V}{T} = k$ where k is a constant

✔ For a given sample of gas at different temperature and volume:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

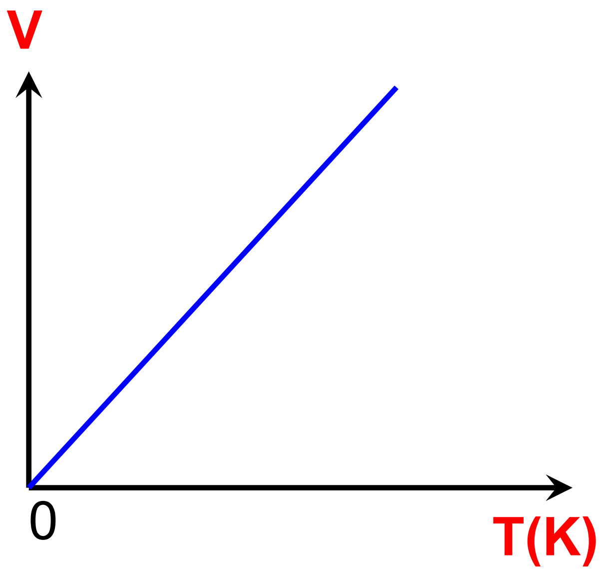
T_1 = initial temperature V_1 = initial volume

T_2 = final temperature V_2 = final volume

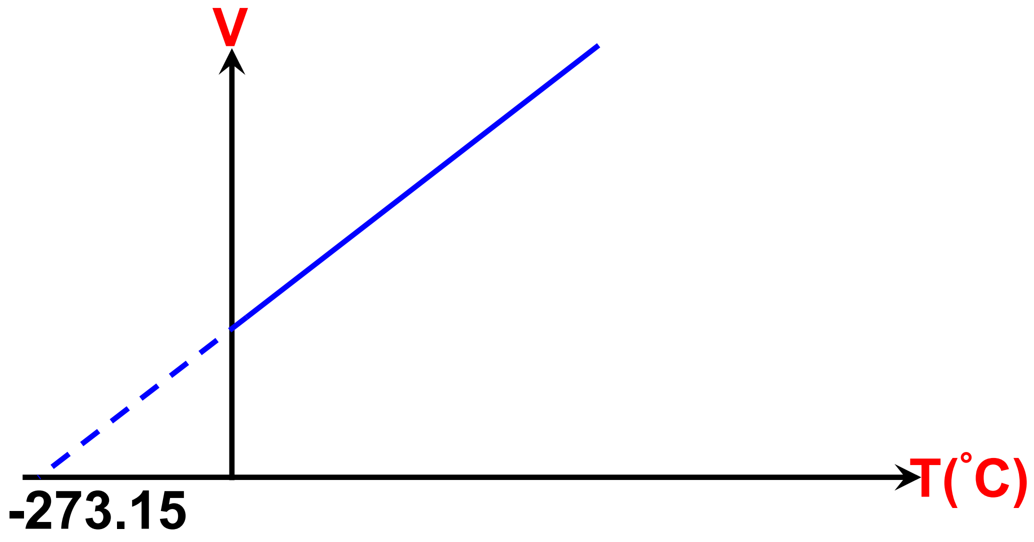
$T(K) = T(^\circ C) + 273$

Charles' Law can be represented by the following graphs at fixed amount of gas and constant pressure:

Volume against temperature (K)



Volume against temperature (°C)



Constant variable :	T & P

Avogadro's Law

by Amedeo Avogadro in 1787

✔ State that for a fixed amount of gas at constant pressure and temperature, gas volume is directly proportional to the number of moles of the gas present.

$$V \propto n \qquad \text{at constant T, P}$$

$$\frac{V}{n} = k \qquad \text{where k is a constant}$$

✔ For a given sample of gas at different mole and volume:

$$\boxed{\frac{V_1}{n_1} = \frac{V_2}{n_2}}$$

n_1 = initial moles
n_2 = final moles

V_1 = initial volume
V_2 = final volume

Constant variable :	n & T

EXAMPLE:

A sample of chlorine gas occupies a volume of 750 mL at a pressure of 5 atm.

Calculate the pressure of the gas if the volume is reduced at constant temperature to 200 mL?

SOLUTION:

V_1 = 750 mL $\qquad$ V_2 = 200 mL

P_1 = 5 atm $\qquad$ P_2 = ?

$$P_1V_1 = P_2V_2$$

$$750 \text{ mL } (5 \text{ atm}) = P_2(200 \text{ mL})$$

$$\mathbf{P_2 = 18.75 \text{ atm}}$$

Constant variable :	n & P

EXAMPLE:

A 452 mL sample of flourine gas is heated from 22 ^oC to187 ^oC at constant pressure. What is its final volume.

SOLUTION:

V_1 = 452 mL V_2 = ?

T_1 = 22^{o}C T_2 = 187^{o}C

= 295 K = 460

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{452\ mL}{295\ K} = \frac{V_2}{460\ K}$$

V_2 = 704.81 mL

Ideal Gas Law

Ideal gas : Gas which obey gas laws and ideal gas equation under all conditions.

obey Boyle's, Charles' and Avogadro's law

combination 3 laws, a single expression relating P,V, n and T is obtained.

Boyle's Law : $V \alpha \frac{1}{P}$ at constant n, T

Charles' Law : $V \alpha T$ at constant n, P

Avogadro's Law : $V \alpha n$ at constant T, P

$$V \alpha \frac{nT}{P}$$

$$V = R\frac{nT}{P}$$

Ideal gas $\longrightarrow$ $\boxed{PV=nRT}$

Where;

P = pressure

V = volume of the gas

n = moles of gas

T = temperature in kelvin

R = gas constant

($8.314\ J\ K^{-1}\ mol^{-1}$ @ $0.08206\ L\ atm\ K^{-1}\ mol^{-1}$)

Units for pressure, volume, universal gas constant and temperature

Pressure, P	**Volume, V**	**Gas constant, R**	**Temperature, T**
Pa @ Nm^{-2}	m^3	8.314 J mol^{-1} K^{-1}	K
atm	L	0.08206 L atm mol^{-1} K^{-1}	K

REMARK: For calculation purpose :**P**, **V**, **R** and **T** must be in the correct unit.

EXAMPLE:

What is the pressure in a 200 cm^3 vessel if 12 g of methane, CH_4 gas is heated to a temperature of 40°C?

SOLUTION:

mass of CH_4 = 12g T = 40^o C = 313 K

V = 200cm^3 = 0.2 dm^3

$$n_{CH_4} = \frac{12g}{16\ gmol^{-1}} = 0.75\ mol$$

PV = nRT

$$P = \frac{(0.75\ mol)(0.08206\ dm^3\ atm\ mol^{-1}K^{-1})(313\ K)}{0.2\ dm^3}$$

= 96.32 atm

NOTE: Density and molar mass of a gas can be calculated by rearranging the Ideal Gas Equation

$$\boxed{PV = nRT}$$

$$PV = \frac{mRT}{M}$$

$$P = \frac{nRT}{V} \quad \text{where } c = \frac{n}{V}$$

$$P = cRT$$

PV mRT = $\frac{\quad}{M}$ where density, $d = \frac{m}{V}$

$$P = \frac{dRT}{M}$$

m = mass of gas

M = molar mass

c = molarity / concentration

d = density

EXAMPLE:

A chemist has synthesized a greenish-yellow compound of chlorine and oxygen and finds that its density is 7.71 g L^{-1} at 36°C and 2.88 atm. Calculate the molar mass of the compound.

SOLUTION:

$d = 7.71 gL^{-1}$ $T = 36^{o}C = 309K$ $P = 2.88$ atm

$$\mathcal{M} = \frac{dRT}{P}$$

$$\mathcal{M} = \frac{(7.71 gL^{-1})(0.08206\ L\ atm\ mol^{-1}K^{-1})(309K)}{2.88\ atm}$$

= 67.88 g mol^{-1}

EXAMPLE:

An unknown gas, **A**, with a mass of 15.7 g at 27°C and 1.5 atm, occupies a volume equivalent to that of 13 g of oxygen at STP. Calculate the molar mass and density of gas **A**.

What would be the volume of the same amount of **A** if pressure were doubled at constant temperature?

SOLUTION:

mass A = 15.7 g T = 27° C = 300 K P = 1.5 atm

$$n_{O_2} = \frac{13}{32\,g\,mol^{-1}}\,g = 0.4063\ mol$$

$V_{O_2} = V_A = 0.4063$ mol x 22.4 L mol^{-1} = 9.1011L

$$d = \frac{m}{V}$$

$$= \frac{15.7g}{9.1011L}$$

$= \mathbf{1.73gL^{-1}}$

$$M = \frac{dRT}{P}$$

$$= \frac{(1.73gL^{-1})(0.08206\ L\ atm\ mol^{-1}K^{-1})(300K)}{1.5\ L}$$

$= \mathbf{28.39\ g\ mol^{-1}}$

EXAMPLE:

An unknown gas, **A**, with a mass of 15.7 g at 27°C and 1.5 atm, occupies a volume equivalent to that of 13 g of oxygen at STP. Calculate the molar mass and density of gas **A**.

What would be the volume of the same amount of **A** if pressure were doubled at constant temperature?

SOLUTION:

$V_1 = 9.1011$ L $P_1 = 1.5$ atm $P_2 = 3.0$ atm

Since **n** and **T** are constant, use Boyle's Law

$$P_1V_1 = P_2V_2$$

$$(1.5 \text{ atm})(9.1011\text{L}) = (3.0 \text{ atm})(V_2)$$

$$\mathbf{V_2 = 4.55\ L}$$

Dalton's Law

The <u>total pressure of mixture of non-reacting gases is the sum of the partial pressures exerted by each of the gas</u> in the mixture

Partial Pressure: Pressure of one component in a mixture of gases

NOTE:

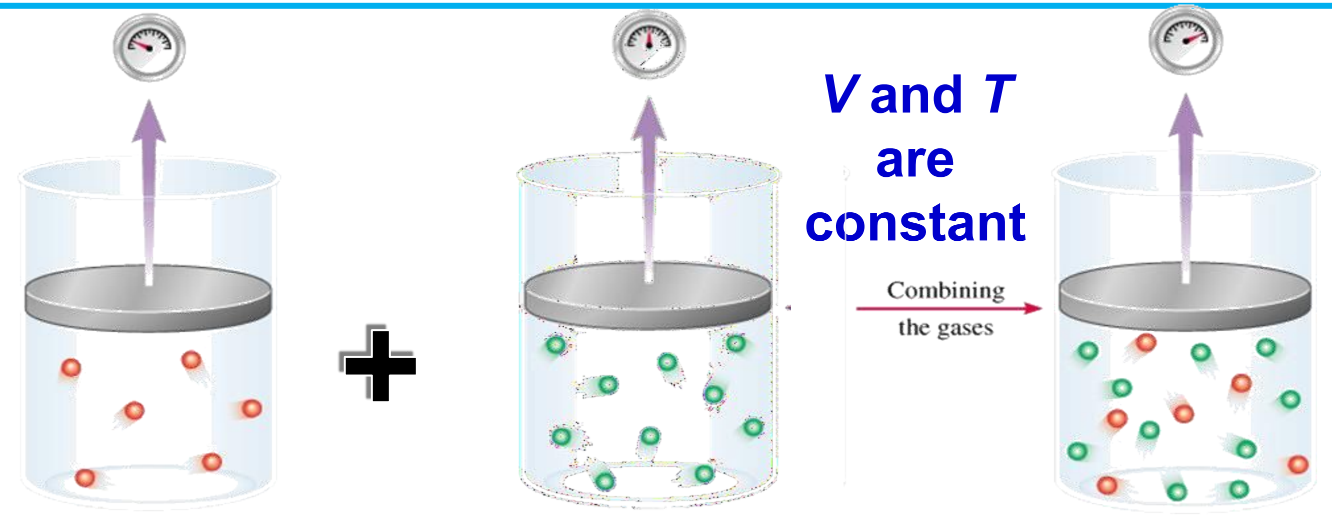
Partial pressure of A is given by:

$$\boxed{P_A = X_A P_T}$$

where X_A = mole fraction of gas A in the mixture

$$X_A = \frac{n_A}{n_T} \Rightarrow \boxed{P_A = \frac{n_A}{n_T} P_T}$$

For a mixture containing gas A and B



Pressure exerted by gas A

Pressure exerted by gas B

Mixture of gases A and B

By using Ideal Gas Equation ;

$$\boxed{P_A = \frac{n_A RT}{V}} \qquad P_B = \frac{n_B RT}{V}$$

$$\boxed{P_T = P_A + P_B}$$

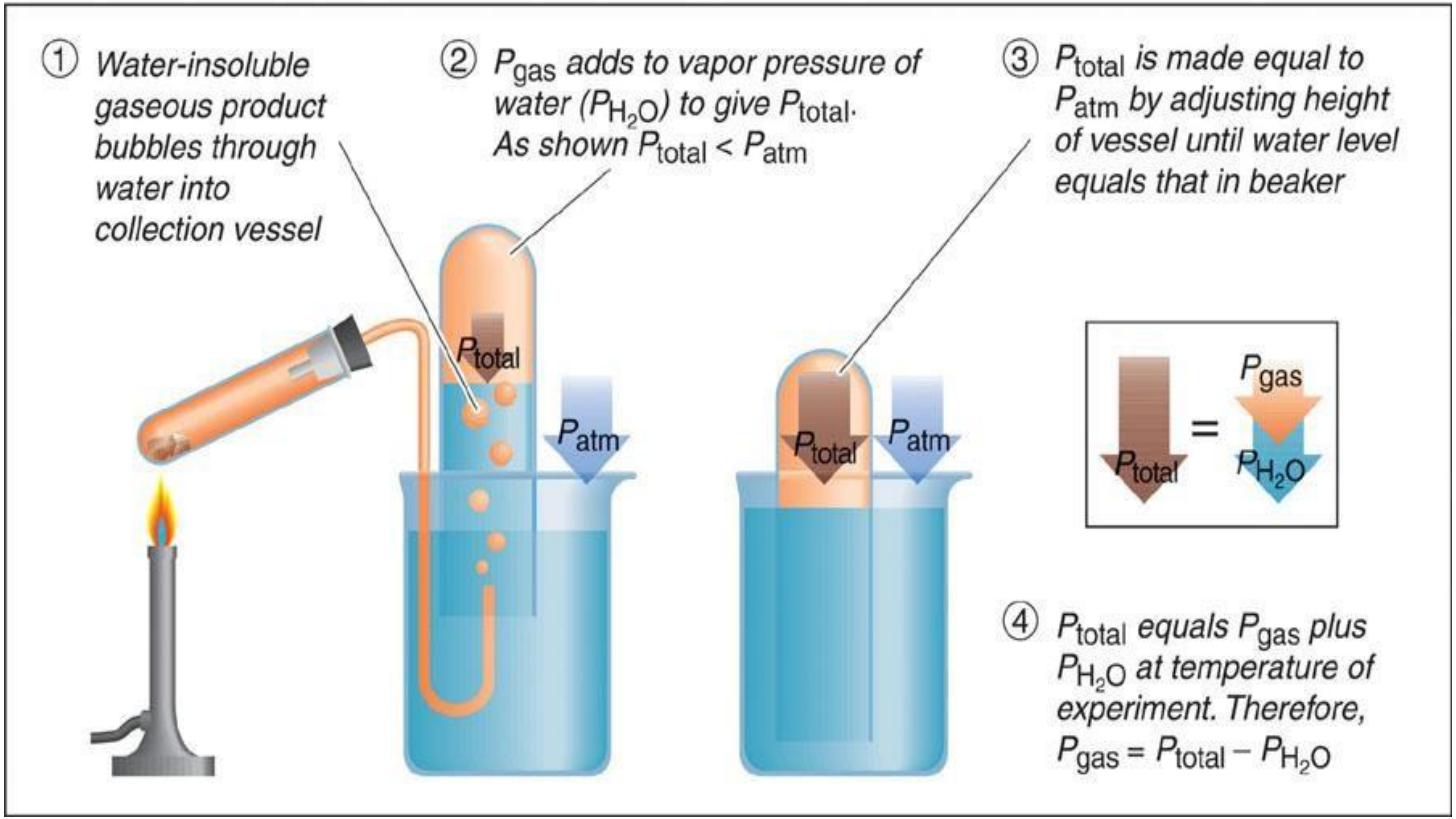
REMINDER: The total mole fraction (X_T) of all gases in the mixture = 1

$$P_T = \frac{n_A RT}{V} + \frac{n_B RT}{V}$$

$$P_T = \frac{n_T RT}{V}$$

Example of Dalton's law in laboratory:

Water Displacement Method – when a gas is collected over water.



- Gas <u>does not</u> react with water
- Total pressure is <u>not only</u> from the particular gas but also from the water vapour.

EXAMPLE:

A mixture of gases contains 0.1 mol of CO_2 and 0.3 mol of O_2. Calculate the partial pressure of the gases if the total pressure is 760 mmHg at a certain temperature.

SOLUTION:

$n_{CO_2} = 0.1 mol$ $n_{O_2} = 0.3 mol$

$P_T = P_{CO_2} + P_{O_2} = 760$ mm Hg

$$P_{CO_2} = \frac{n_{CO_2}}{n_T} P_T$$

$$= \left(\frac{0.1}{0.1 + 0.3}\right) 760$$

= 190 mm Hg

$P_{O_2} = P_T - P_{CO_2}$

$= 760 - 190$

= 570 mm Hg

EXAMPLE:

A sample of gas at 5.88 atm contains 1.2 g CH_4, 0.4 g H_2 and 0.1 g He. Calculate the partial pressure of CH_4 in the mixture.

SOLUTION:

mass CH_4 = 1.2g mass H_2 = 0.4 g mass He = 0.1 g

$P_T = P_{CH_4} + P_{H_2} + P_{He} = 5.88$ atm

$n_{CH_4} = \frac{1.2\text{ g}}{16\text{ gmol}^{-1}} = 0.075$ mol

$n_{H_2} = \frac{0.4\text{ g}}{2\text{ gmol}^{-1}} = 0.2$ mol

$n_{He} = \frac{0.1\text{ g}}{4\text{ gmol}^{-1}} = 0.025$ mol

$n_T = n_{CH_4} + n_{H_2} + n_{He}$

$= 0.075\text{ mol} + 0.2\text{ mol} + 0.025\text{ mol} = 0.3$ mol

$P_{CH_4} = \frac{n_{CH_4}}{n_T} P_T$

$= \left(\frac{0.075\text{ mol}}{0.3\text{ mol}}\right)(5.88\text{ atm})$

= 1.47 atm

EXAMPLE:

Consider the reaction below :

$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$

A sample of 5.45 L of oxygen is collected over water at a total pressure of 735.5 torr at 25°C. How many grams of oxygen have been collected? (at 25°C, the vapour pressure of water is 23.8 torr)

SOLUTION:

P_T = 735.5 torr V = 5.45 L T = 298 K

$P_T = P_{O_2} + P_{H_2O}$

P_{O_2} = 735.5 torr - 23.8 torr

= 711.7 torr

$\frac{711.7}{760} = 0.9364$ atm

$P_{O_2}V = n_{O_2}RT$

$n_{O_2} = \frac{(0.9364\ \text{atm})(5.45\ \text{L})}{(0.08206\ \text{L atm mol}^{-1}\text{K}^{-1})(298\ \text{K})}$

= 0.2087 mol

mass of O_2 = 0.2087 mol x 32 g mol^{-1}

= **6.68g**

REAL GASES : DEVIATION FROM IDEAL BEHAVIOUR

- Gases which **does not** obeys ideal gas equation (also known as non-ideal gas)
- Real gas deviates from ideal gas behaviour because:

n1 The gas molecules have a certain volume.

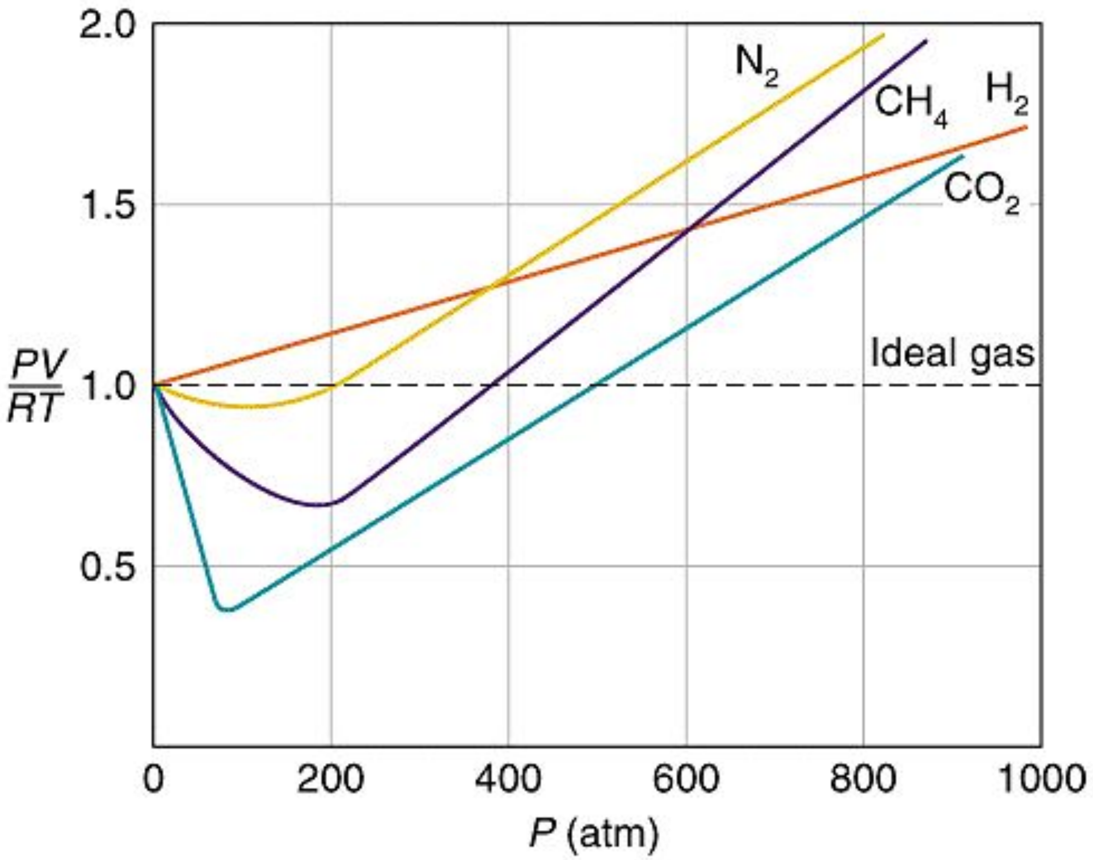
n2 There are intermolecular attractive forces between the gas molecules

- The gas obeys **van der Waals Equation.**

- Deviations occurs at:

1. High Pressure
2. Low Temperature

REAL GASES : DEVIATION FROM IDEAL BEHAVIOUR

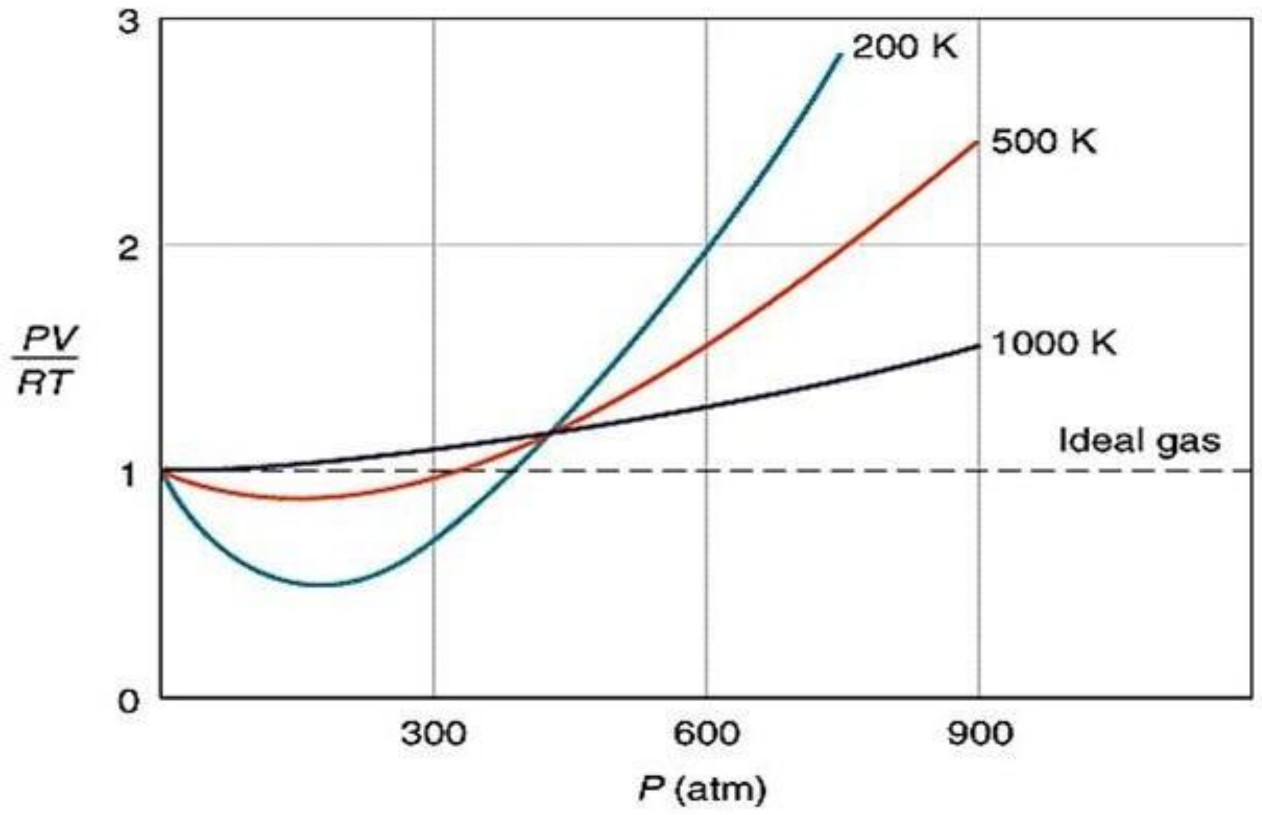


Deviation of real gas from ideal gas behaviour can be shown by plotting PV/RT against pressure.

The graph shows :

- ✔ The gas behaves almost ideally at very low pressure & deviate from ideal behaviour when pressure increases.

REAL GASES : DEVIATION FROM IDEAL BEHAVIOUR



- Deviation of real gas from ideal gas behaviour can be shown by plotting PV/RT against pressure (at different temperatures)

The graph shows :

- The gas behaves almost ideally at very high temperature & deviate from ideal behaviour when temperature decreases.

REAL GASES : DEVIATION FROM IDEAL BEHAVIOUR

Reason of the deviation :

1 High pressure

- The volume of container is reduced, the distance between gas molecules getting closer.
- The intermolecular attractive forces now become significant.
- Gas molecules now occupy sizeable portion of the container

2 Low Temperature

- The kinetic energy of the gas molecules become smaller.
- The intermolecular attractive forces become significant.

The real gas obeys van der Waals equation

Since real gas does not exhibit ideal gas behavior at high pressure and low temperature, the ideal gas equation (PV=nRT) needs to be adjusted.

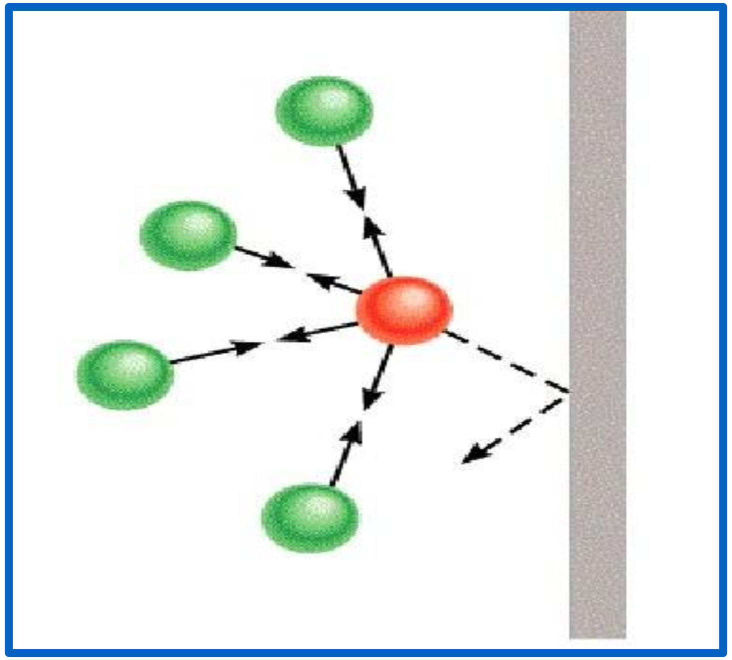
Adjusting the equation, two parameters need to be reconsidered:

a) Intermolecular attractive forces between the gas molecules.

b) volume of the gas molecules.

REAL GASES : van der Waals Equation

CORRECTION ON PRESSURE



The intermolecular attractive forces acting on molecule:

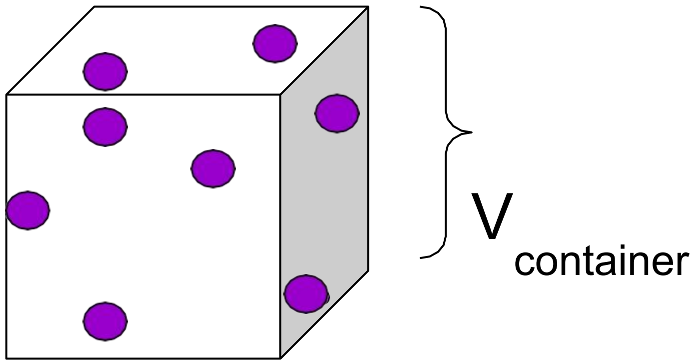
- make the molecules move slower
- give less impact to the wall
- pressure exerted by the real gas is less compared to the ideal gas

$$\boxed{P_{real} < P_{ideal}}$$

- the term pressure need to be corrected by adding $\frac{n^2 a}{V^2}$,therefore :

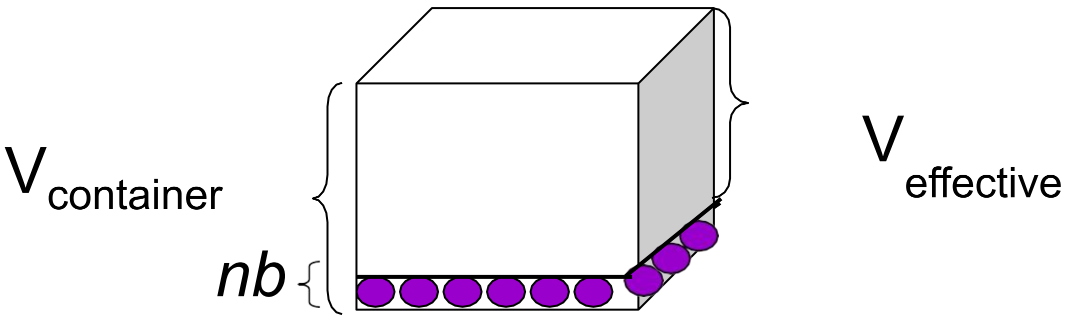
$$\boxed{P_{ideal} = P_{real} + \frac{n^2 a}{V^2}}$$

CORRECTION ON VOLUME



In ideal gas equation,

$V = V_{container}$



- However, each molecule in a real gas occupy a finite volume, therefore a real gas has a larger volume than ideal gas.
- The space occupied by molecules restrict the movement of the gas molecules. Hence must be subtracted from the total volume.
- So, the effective volume of the gas is not V (volume of container) but becomes $V - nb$.

$$\boxed{V_{effective} = V_{container} - nb}$$

n: number of moles of the gas

b : a constant

nb: volume occupied by *n* moles of the gas

Combining both factors into an ideal gas equation :

$$PV = nRT \Rightarrow \boxed{\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT}$$

van der Waals equation

- ✔ Value of ***a*** indicate how strongly molecules of gas attract each other.
- ✔ The stronger the intermolecular attractive forces between gas molecules, the higher *a* value.

- ✔ Value of ***b*** indicate the molecular size of gas molecules.
- ✔ The larger the the size of molecule (or atom), the higher *b* value.

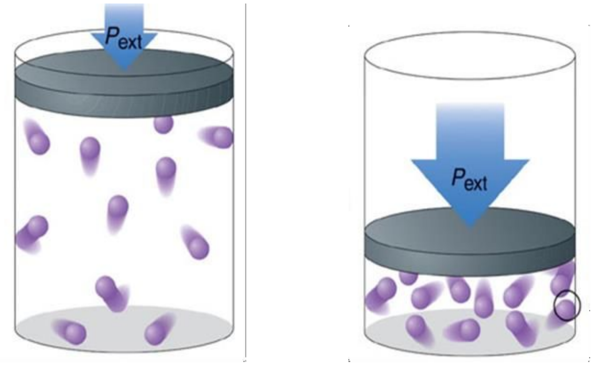
EXAMPLE

GAS	*a* (L^2 atm mol^{-2})	b (L mol^{-1})
He	0.034	0.0237
CO_2	3.59	0.0427

- Helium atoms have a smaller *a* value because they have weaker intermolecular attractive forces for one another than CO_2
- Helium atoms have a smaller *b* value because their atomic size are smaller than CO_2

REAL GASES BEHAVES ALMOST IDEALLY UNDER TWO CONDITIONS:

1 Low pressure



Low pressure **high pressure**

- As the volume of container increases:
- The molecules in a gas are far apart, hence the intermolecular attractive forces are negligible.
- The volume of the molecules is almost zero compared to the total volume & can be neglected

2 High Temperature

- The gas molecules possess high kinetic energy.
- This allows them free themselves from the intermolecular attractive forces. The particles have enough energy to overcome intermolecular forces.
- The intermolecular forces canbe neglected.

EXAMPLE:

The van der Waals equation for 1 mol of real gas is as follow :

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

Determine and explain whether :

a) H_2 or H_2S molecules has a higher of a and b.

b) a or b is related to boiling point.

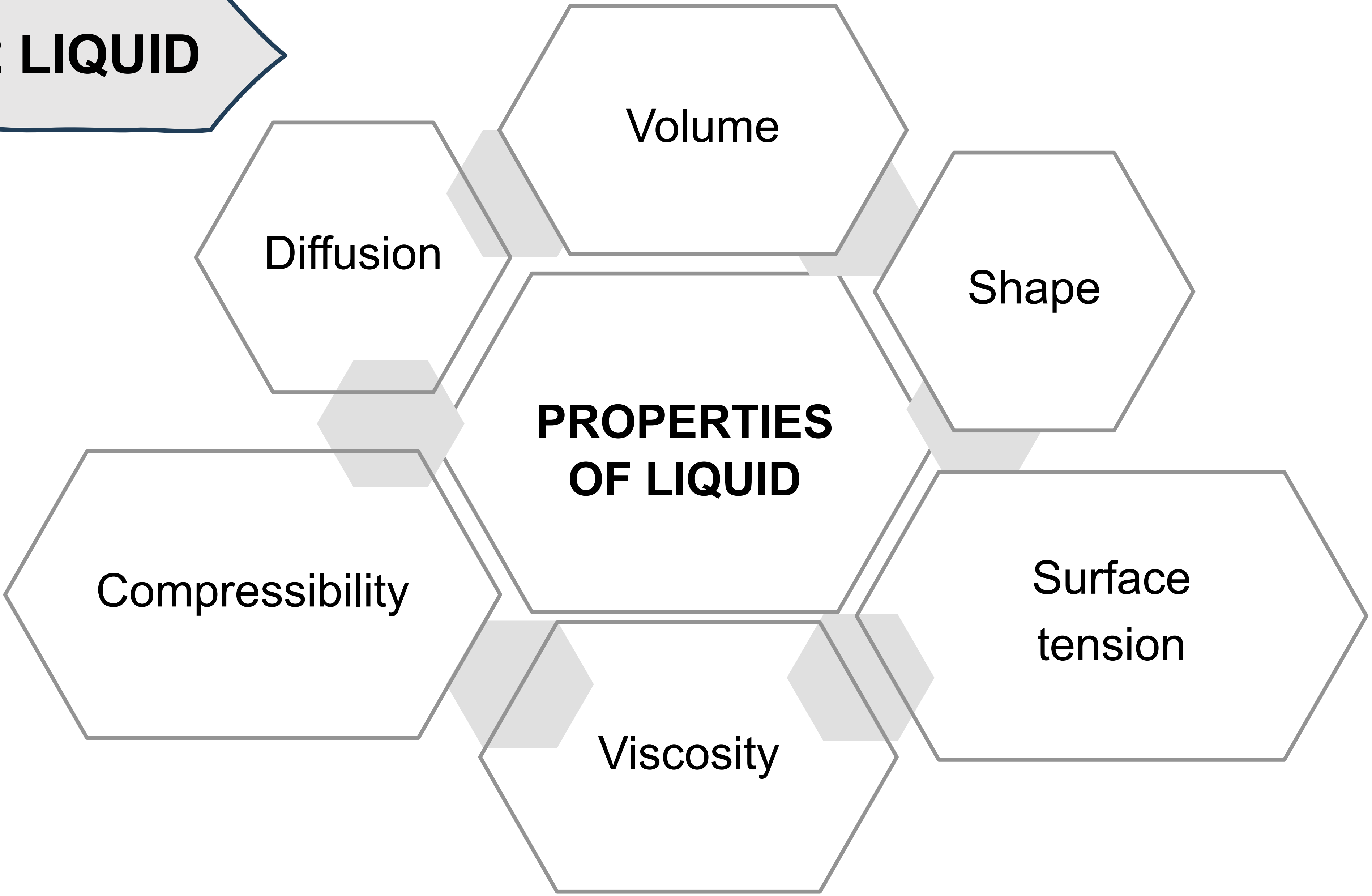
SOLUTION:

a) H_2S molecules have a higher *a* value because they have the stronger intermolecular forces.

H_2S molecules have a higher *b* value because their size are bigger, therefore have a higher volume.

b) *a* is related to boiling point because stronger intermolecular forces, thus more energy required to overcome the intermolecular forces, higher boiling point.

5.2 LIQUID



PROPERTIES OF LIQUID
Volume
Shape
Surface tension
Viscosity
Compressibility
Diffusion

1 Volume

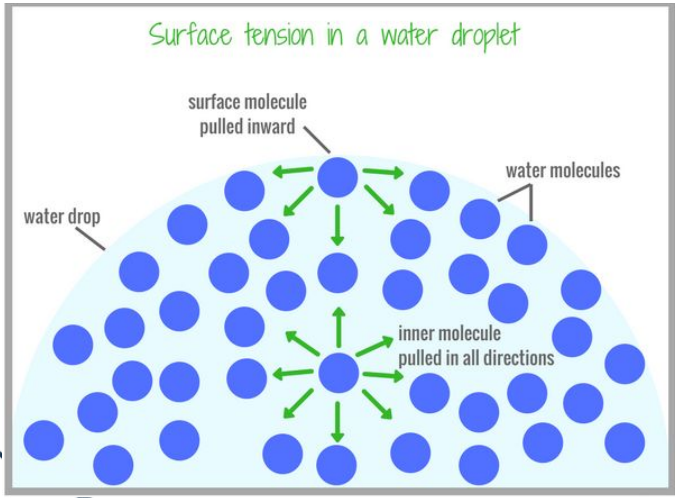
- **A liquid has a definite volume.**
- Intermolecular attractive forces are strong enough to hold molecules close together to retain the volume of liquid

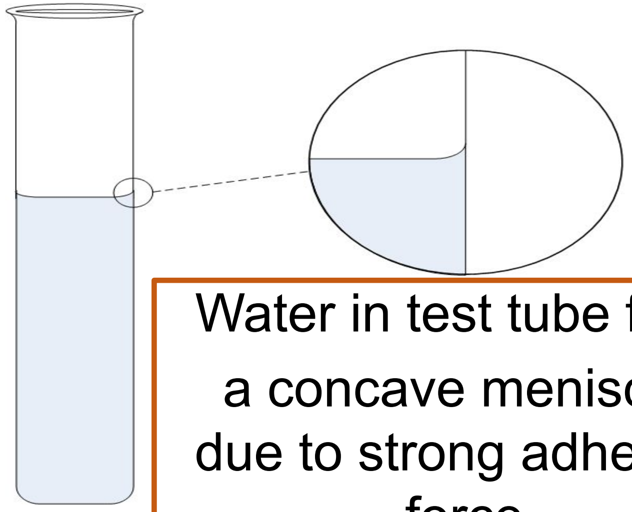
2 Shape

- **Take the shape of container**.
- Intermolecular attractive forces are not strong enough to hold the molecules in fixed position.
- The particles are able to move freely causes it to take the shape of the container

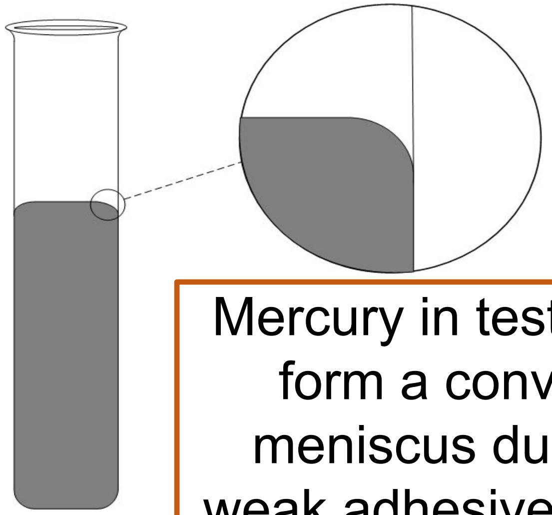
3 Surface tension

- The amount of energy required to stretch or increase the surface of a liquid by a unit area.
- Caused by the intermolecular attractive forces acting within the liquid molecules
- Molecules at the surface are pulled inwards and sideways from the neighboring molecules.
- The inner pulls on the surface molecules cause the liquid to minimise its surface area.
- Hence, the stronger the intermolecular attractive force, the higher the surface tension.





Water in test tube form a concave meniscus due to strong adhesive force



Mercury in test tube form a convex meniscus due to weak adhesive force

4 Viscosity

- Is the tendency of liquid to resist flow.
- The greater the viscosity, the harder/slower it flows
- Factors affecting the viscosity of a liquid:
  i. Size of molecules
  ii. Intermolecular attractive forces acting between molecules
  iii. Temperature of the liquid

- The viscosity of the liquid increases if

Size of molecules	high
Strength of Intermolecular attractive forces of molecules	high
Temperature	low

Factors Affecting the Viscosity of Liquid

Size of Molecules

- As the molecular size increases (bigger molecular weight), the resistance to flow increases (harder to flow).
- Viscosity of the liquid increases.

Strength of Intermolecular Attractive Forces

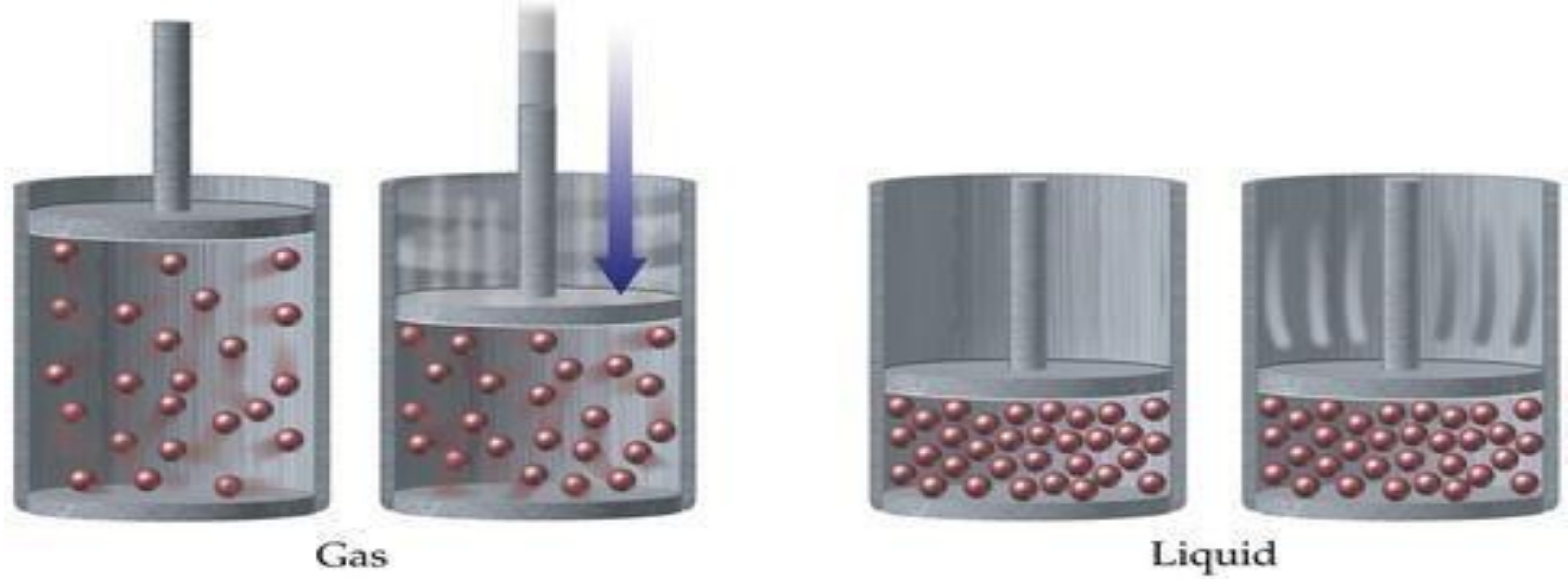
- As the strength of intermolecular attractive forces increases, the resistance to flow increases(harder to flow)
- Viscosity of the liquid increases

Temperature

- As the temperature increases, the average kinetic energy of molecules to overcome the intermolecular forces increases.
- Resistance to flow decreases, viscosity of the liquid decreases

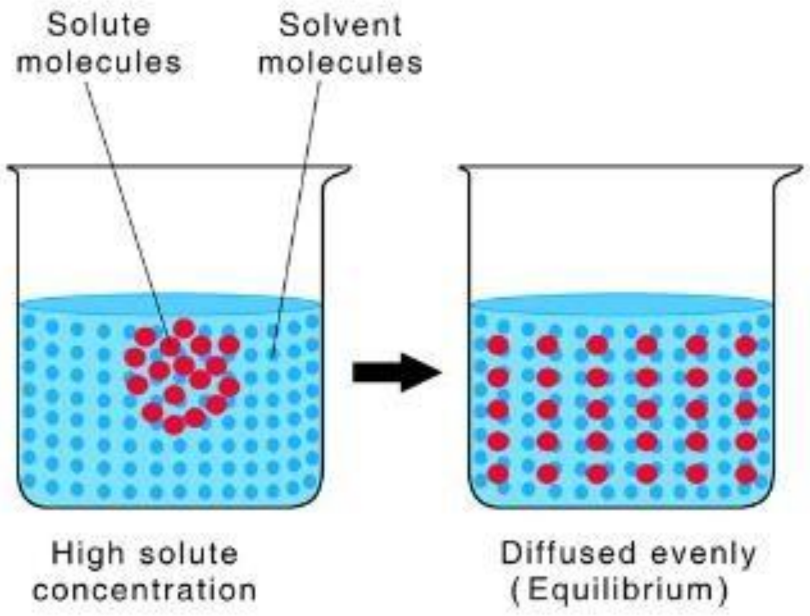
5 Compressibility

- In liquid, the particles are packed closely together.
- There is very little empty space between the molecules hence liquid is more difficult to compress than gas.



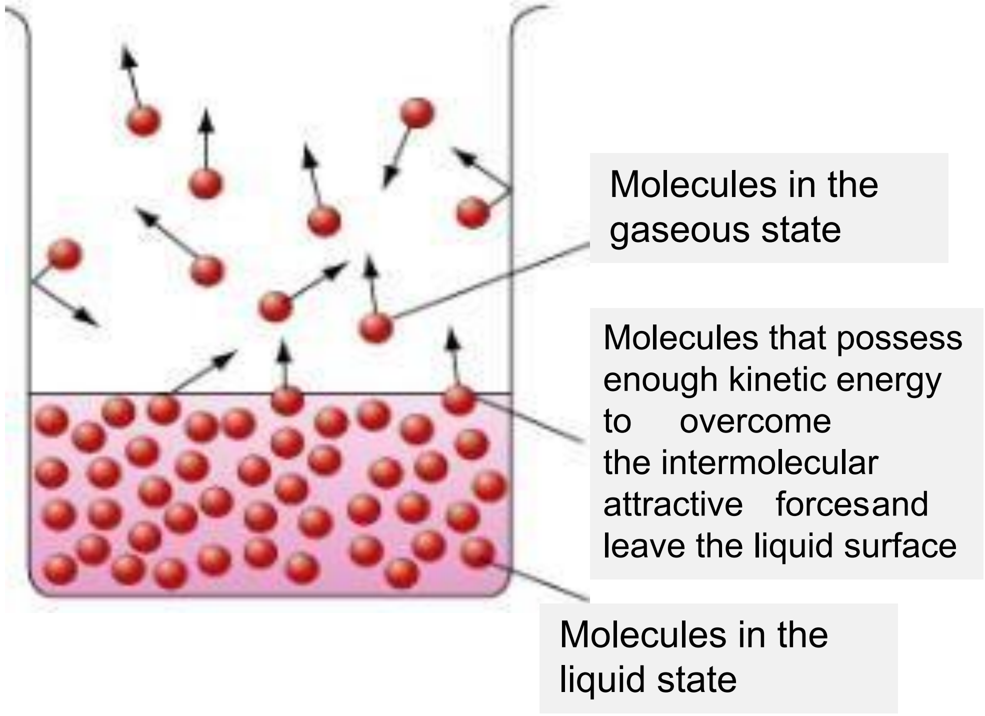
6 Diffusion

- Diffusion is the movement of particles from a region of higher concentration to a region of lower concentration.
- Diffusion rate of liquid is much slower than gases , but faster than solid because:
  - a) Molecules of liquid are closely packed.
  - b) Stronger intermolecular attractive forces restrict the movement of the liquid molecul



VAPORISATION

- Vaporisation is a process of changing state from liquid to vapour.
- Molecules in a liquid move freely with different magnitude kinetic energy.
   - Higher kinetic energy, moves faster
   - Lower kinetic energy, moves slower
- Some molecules have higher kinetic energy than some other molecules.
- Molecules at the surface that possess enough kinetic energy to overcome the intermolecular attractive forces can escape as vapour in the gas phase.



FACTORS THAT AFFECT VAPORISATION

Surface Area

- Higher surface area, more liquid molecules are on the liquid surface.
- More molecules are able to escape into vapour
- Vaporisation rate increases

Temperature

- The higher temperature, the higher kinetic energy
- More liquid molecules are able to overcome the intermolecular forces and escape into vapour phase
- Vaporisation rate increases

Strength of intermolecular forces

- The lower intermolecular force, the higher the vaporisation rate

CONDENSATION

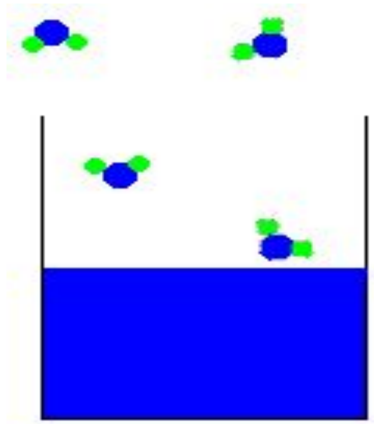
- Condensation is a process of changing state from vapour to liquid.
- It occurs when the vapour molecules are cooled or when pressure is exerted.
- The molecules will lose their kinetic energy during collision. Under this condition, the molecules move slowly and closer to each other.
- The intermolecular attractive forces are strong enough to pull the molecules together and trapped among the liquid molecules hence, liquid is formed.

Vapour molecules lose kinetic energy and trapped among liquid molecules

VAPOUR MOLECULES & DYNAMIC EQUILIBRIUM

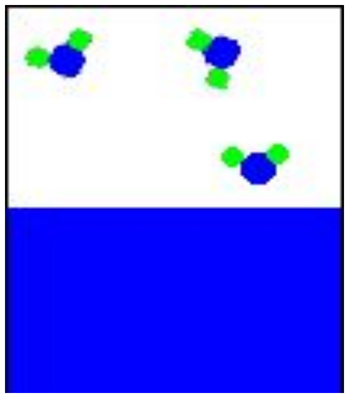
- Vapour molecules: molecules that escaped from the surface of a liquid.
- If evaporation occurs in an open system, vapour molecules will diffuse away until the liquid dries up.
- If evaporation occurs in a closed system, vapour molecules are trapped in the container.
- When the rate of vaporisation is equal to the rate of condensation, a state of dynamic equilibrium is reached.
- The pressure exerted at this stage is called equilibrium vapour pressure of the liquid.

Vapour molecules will diffuse away

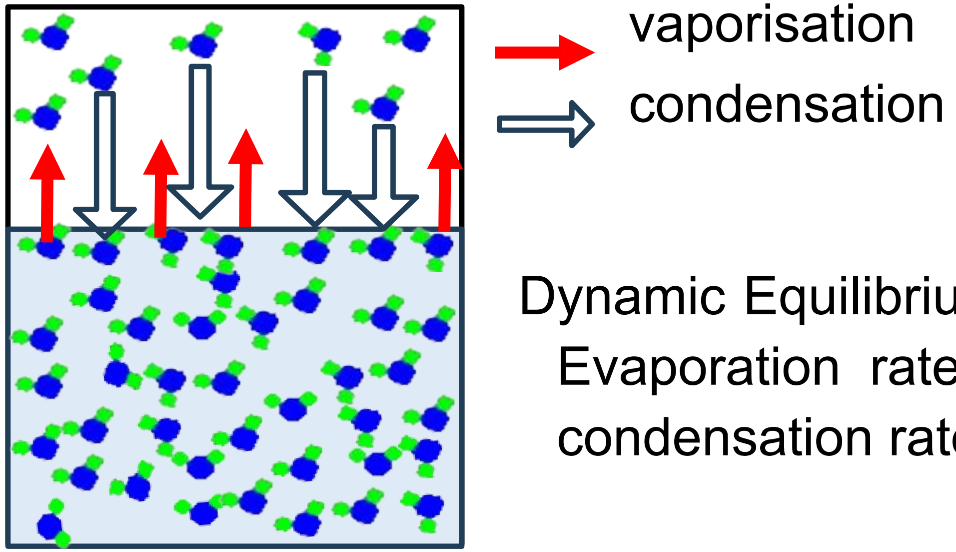


Open system

Vapour molecules are trapped in the container



Closed system



Dynamic Equilibrium:
Evaporation rate = condensation rate

VAPOUR PRESSURE

Vapour pressure is the **pressure exerted** by vapour in equilibrium with its liquid in a **closed container.**

Factors Affecting Vapour Pressure

Intermolecular attractive forces

- At constant temperature, molecules with weak intermolecular attractive forces will have higher vapour pressure.
- Hence, easier to escape from the liquid surface and more volatile.

Temperature

- As temperature increases, molecules will have higher kinetic energy.
- More molecules will overcome the intermolecular attractive forces and escape from the liquid surface to form vapour.
- Hence, the vapour pressure increase.

BOILING POINT

- Boiling Point : The temperature at which the vapour pressure of a liquid equals the atmospheric or external pressure.
- Boiling occurs at the surface and inner part of the liquid at specific temperature and pressure.
- During the boiling process, temperature remain constant even heat is supplied.
- Normal boiling point is the temperature at which a liquid boils when the external pressure is 1 atm.

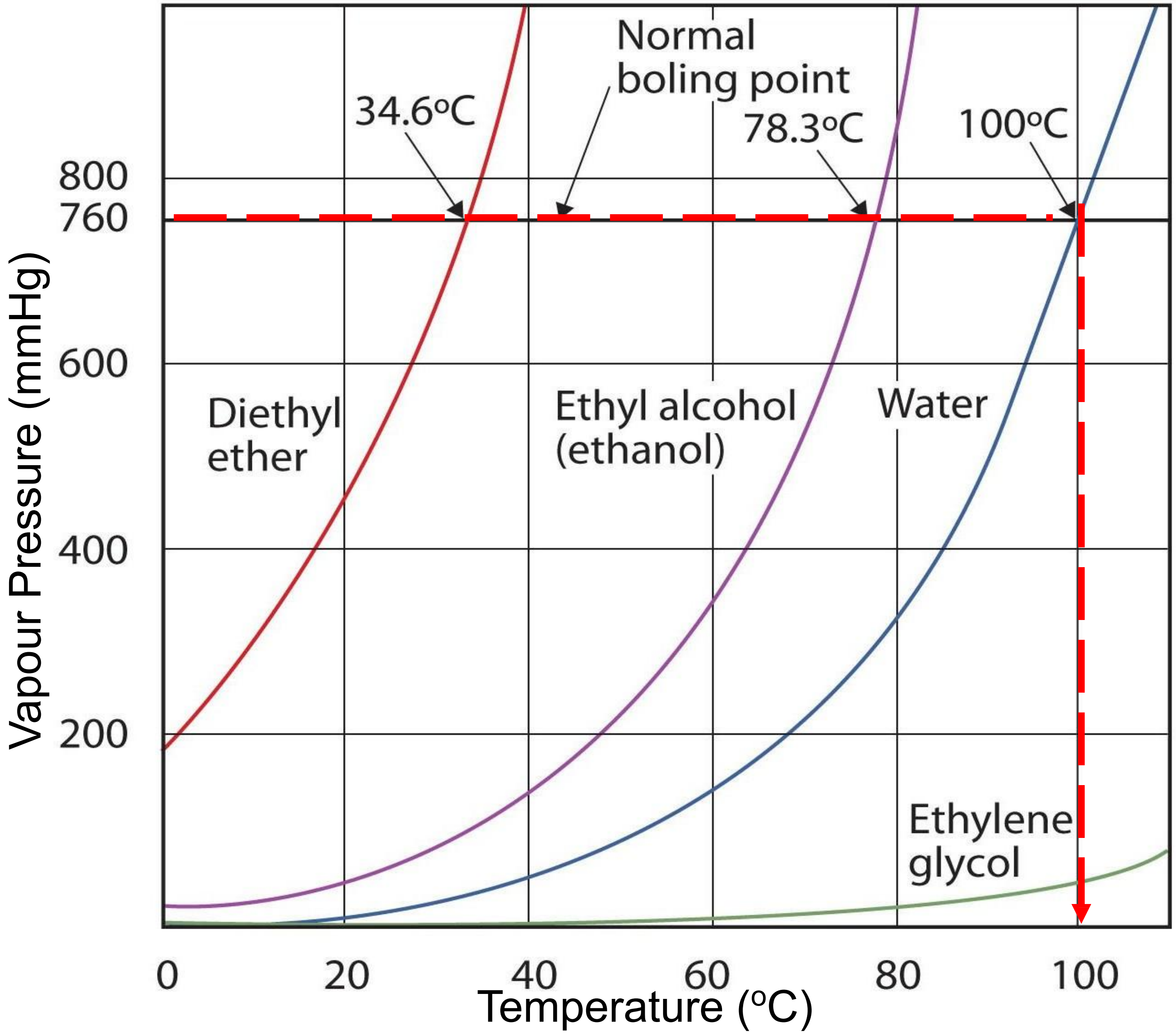
Factors Affecting Boiling point

Atmospheric pressure

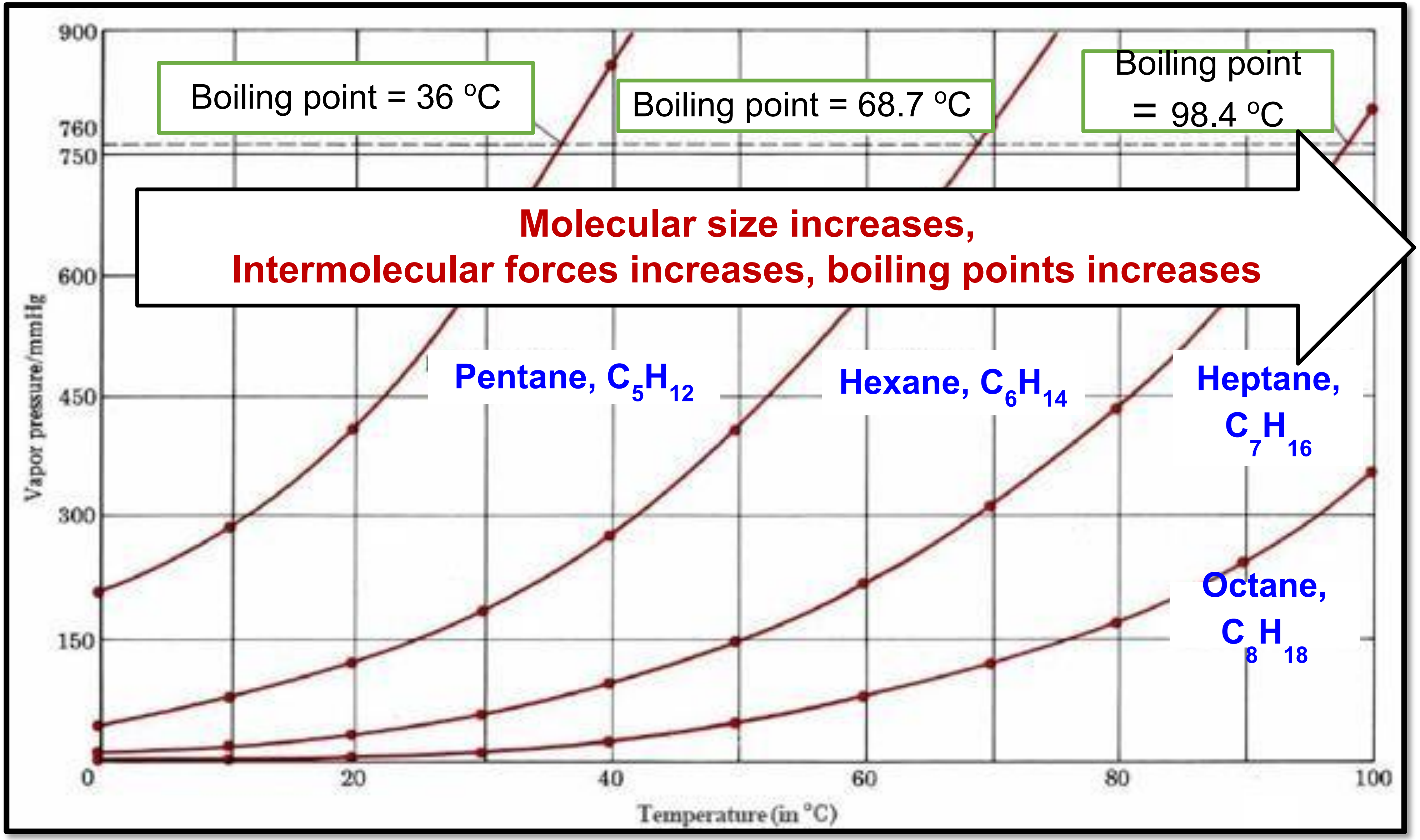
- The higher the atmospheric pressure, the higher the boiling point of a liquid.
- More heat is needed so that the molecules have more kinetic energy to vapourise and to reach vapour pressure equal to external atmospheric pressure

Intermolecular forces

- Liquid with stronger intermolecular attractive forces will have lower vapour pressure and higher boiling points



Normal
boling point
34.6°C
78.3°C
100°C
800
760
600
400
200
Vapour Pressure (mmHg)
Diethyl
ether
Ethyl alcohol
(ethanol)
Water
Ethylene
glycol
0
20
40
60
80
100
Temperature (°C)


Boiling point = 36 °C
Boiling point = 68.7 °C
Boiling point = 98.4 °C
Molecular size increases,
Intermolecular forces increases, boiling points increases
Pentane, C_5H_{12}
Hexane, C_6H_{14}
Heptane, C_7H_{16}
Octane, C_8H_{18}
Vapor pressure/mmHg
900
760
750
600
450
300
150
0
20
40
60
80
100
Temperature (in °C)

EXAMPLE:

a) Define boiling point.

b) "The boiling point of a liquid increases with the increase in the strength of intermolecular forces". Explain the statement.

SOLUTION:

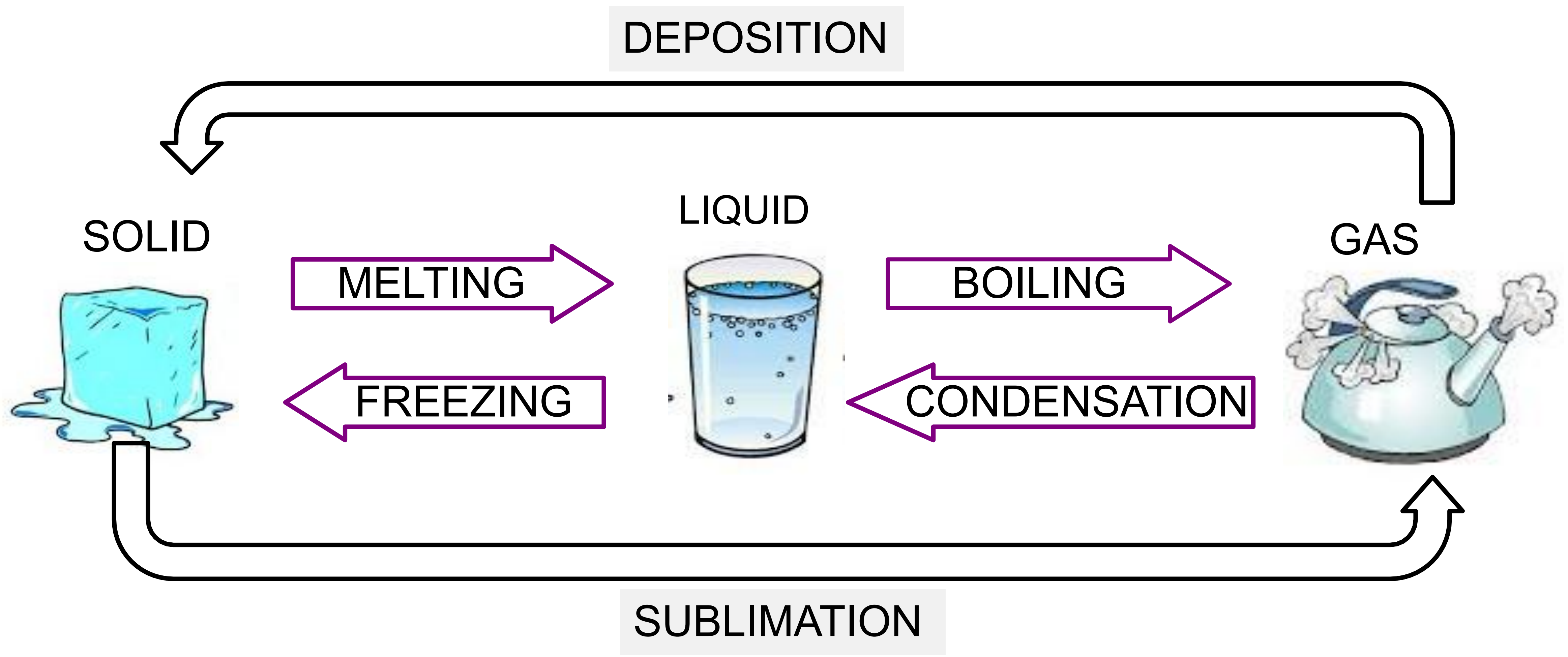
a) The temperature at which the vapour pressure of a liquid equals the atmospheric or external pressure

b) Liquid whose intermolecular attractive forces are stronger will have higher boiling point.
This is because more energy is needed to overcome the intermolecular attractive forces acting on them, in order to vapourise and reach a vapour pressure that is equivalent to the atmospheric pressure.

5.3 SOLID

Strength Of Interparticle Forces	Strongenough to hold solid particles together firmly in place
Arrangement Of Particles	Closely packed and held in rigid and orderly arrangement
Molecular Motion	Cannot move freely but can only vibrate and rotate about fixed position
Shape And Volume	Definite shape and fixed volume
Compressibility	Almost none : very difficult to be compressed
Density	High

In Principle, Solid, Liquid And Gas States Are Inter- Convertible



FREEZING (SOLIDIFICATION)

- Freezing is a process of changing state liquid to solid phase.
- When heat from a liquid is removed (liquid is cooled), the kinetic energy of the particles is decreased.
- A point is reached when the particles can no longer overcome the intermolecular attractive forces and move more slowly and finally arranges into fixed positions to become solid

MELTING (FUSION)

- Melting is a process of changing state solid to liquid phase.
- When molecules in the solid state are heated, they gain kinetic energy and vibrate faster.
- When the kinetic energy of the molecules is strong enough to overcome the intermolecular attractive forces, the molecules move freely and no longer confined to fixed position, thus solid converts to liquid.

SUBLIMATION

- Sublimation is a process of changing state solid to gas.
- Occurs when molecules on the surface of a solid possess enough kinetic energy to overcome the intermolecular attractive forces leave the solid phase as vapour.

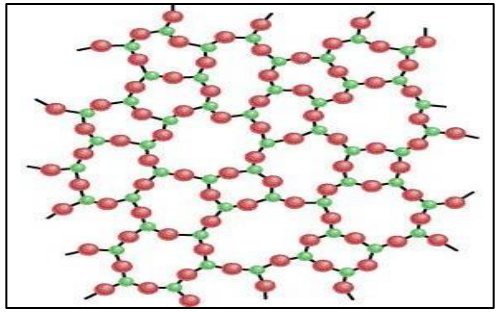
DEPOSITION

- Deposition is a process of changing state gas to solid.
- When cooled under certain conditions vapour molecules lose kinetic energy and can no longer overcome the intermolecular attractive forces thus molecules arrange into fixed positions to become solid.

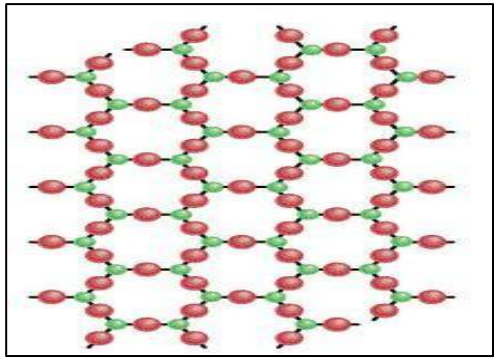
TYPES OF SOLID

Solid can be divided into two types :

AMORPHOUS	CRYSTALLINE
Particles are in irregular @ disorderly random arrangement.	Particles are in regular arrangement @ orderly
No definite melting point, melting occurs over a temperature range	Sharp @ well defined melting point.
Shatter irregularly	Shatter to crystalline shaped pieces
Eg : glass, plastic material, rubber	Eg : sugar, ice, salt



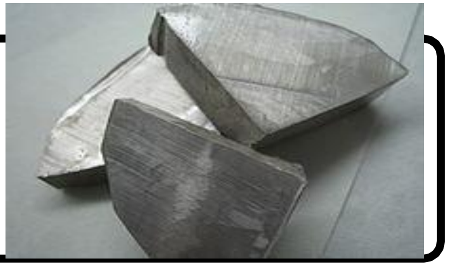
Amorphous Solid : glass



Crystalline Solid : SiO_2

Types of Crystalline Solid

METALLIC SOLID



examples : all metallic elements : Na, Mg, Fe

IONIC SOLID

examples : $MgCl_2$, KCl

MOLECULAR COVALENT SOLID

example:

element: iodine (I_2), sulphur (S_8), phosphorus (P_4)

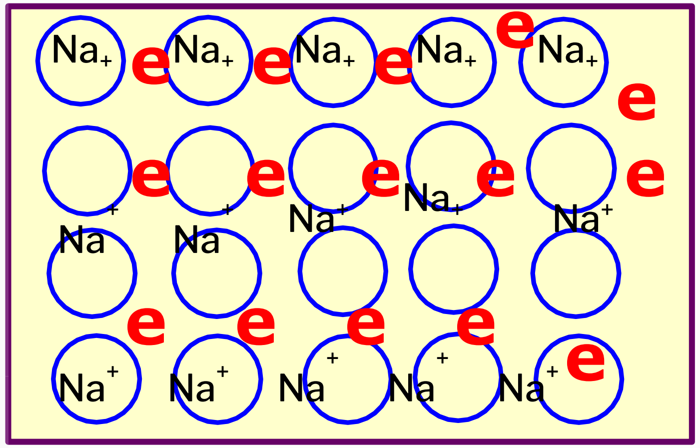
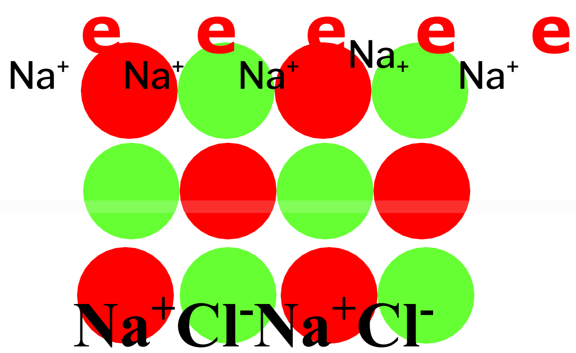
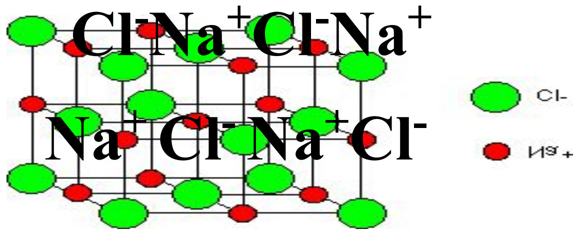
compound: dry ice (solid CO_2), (H_2O),

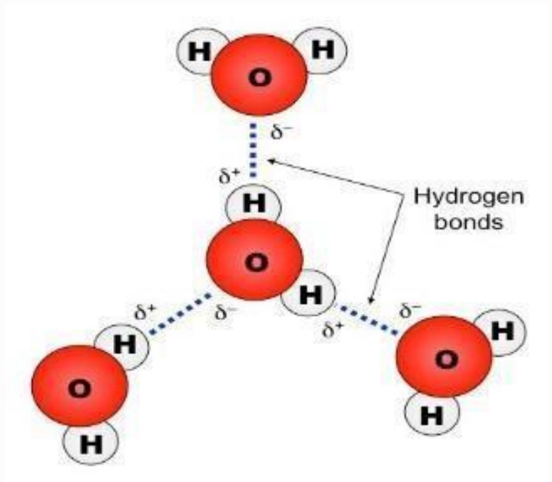
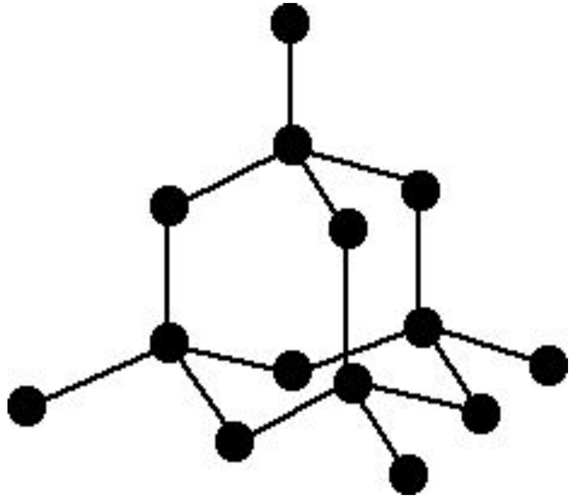
GIANT COVALENT SOLID

example:

element: examples : element→ diamond, graphite, Si

compound → silicon(IV) oxide, SiO_2

CRYSTALLINE SOLIDS	TYPE OF BONDING	EXAMPLE
Metallic Solid (Na, Mg, Fe, Al)	**Metallic bond** Held together by the electrostatic force of attraction between the positively charged ions and the negatively charged of sea of delocalised valence electrons	
Ionic Solid (NaCl,LiF, MgO)	**Ionic Bond** Held by strong electrostatic force of attraction between positive and negative ions	 

CRYSTALLINE SOLIDS	TYPE OF BONDING	EXAMPLE
Molecular Covalent Solid (CO_2 (dry ice), I_2,	**Dispersion forces, dipole-dipole interaction, hydrogen bond** Held together by intermolecular forces such as van der Waals forces or hydrogen bonds	
Giant Covalent Solid C (diamond), SiO_2 (quartz)	**Covalent bond** Giant/ infinite/ network covalent solid in which the solid bound by strong covalent bond.	

EXAMPLE:

a) Define melting point and normal melting point of a substance.

b) Describe the melting process in terms of the molecular kinetic theory.

SOLUTION:

a) Melting point: Temperature when solid changes to liquid
Normal melting point: The temperature at which the solid and liquid are in equilibrium at total pressure of 1 atm.

b) When molecules in solid are heated, molecules gain kinetic energy and vibrate faster. When kinetic energy of the molecules is strong enough to overcome the attractive forces, the molecules moves freely and are no longer confined to fixed position, thus solid converts to liquid.

5.4 PHASE DIAGRAM

- Substance may exist as solid, liquid, gas state, at different temperature and pressure
- When temperature and/or pressure are changed, substance will be transformed from one state to another.
- However, at certain temperature & pressure substance may exist in 2 @ 3 different states at the same time.
- Under this condition, substance is said to be in a phase equilibrium ie equilibrium between different phases in a system

PHASE

- Is a homogenous part of the system in contact with other parts of the system but separated from them by a well-defined boundary

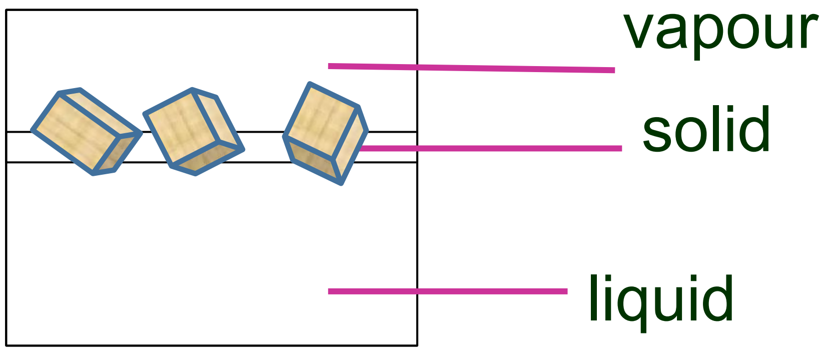
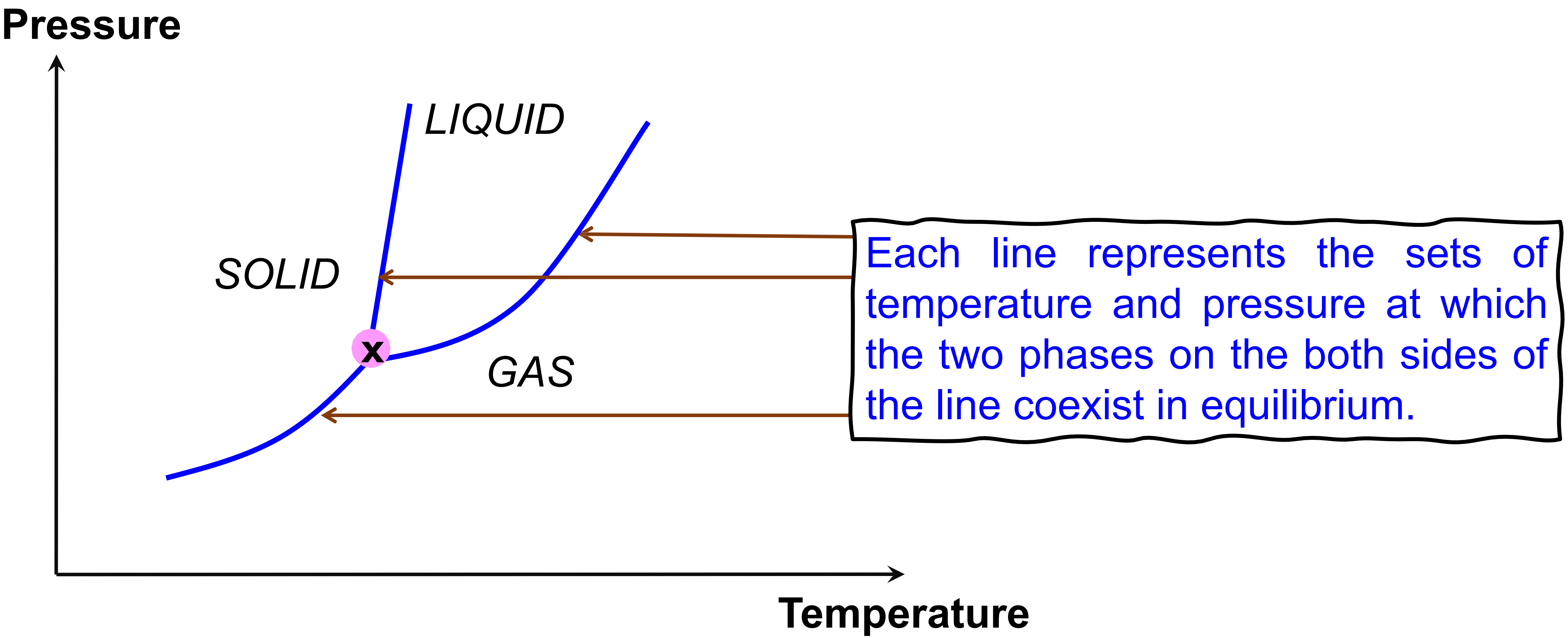


Figure shows the system that consist of ice, water and vapour

PHASE DIAGRAM

Graphical representation of the phase changes of a substance at various conditions of temperature and pressure.



Line

- Represents the boundary between two phases.
- Two phases can exist together at equilibrium

Area(region)

- Represents the conditions under which the substance will be in that particular phase

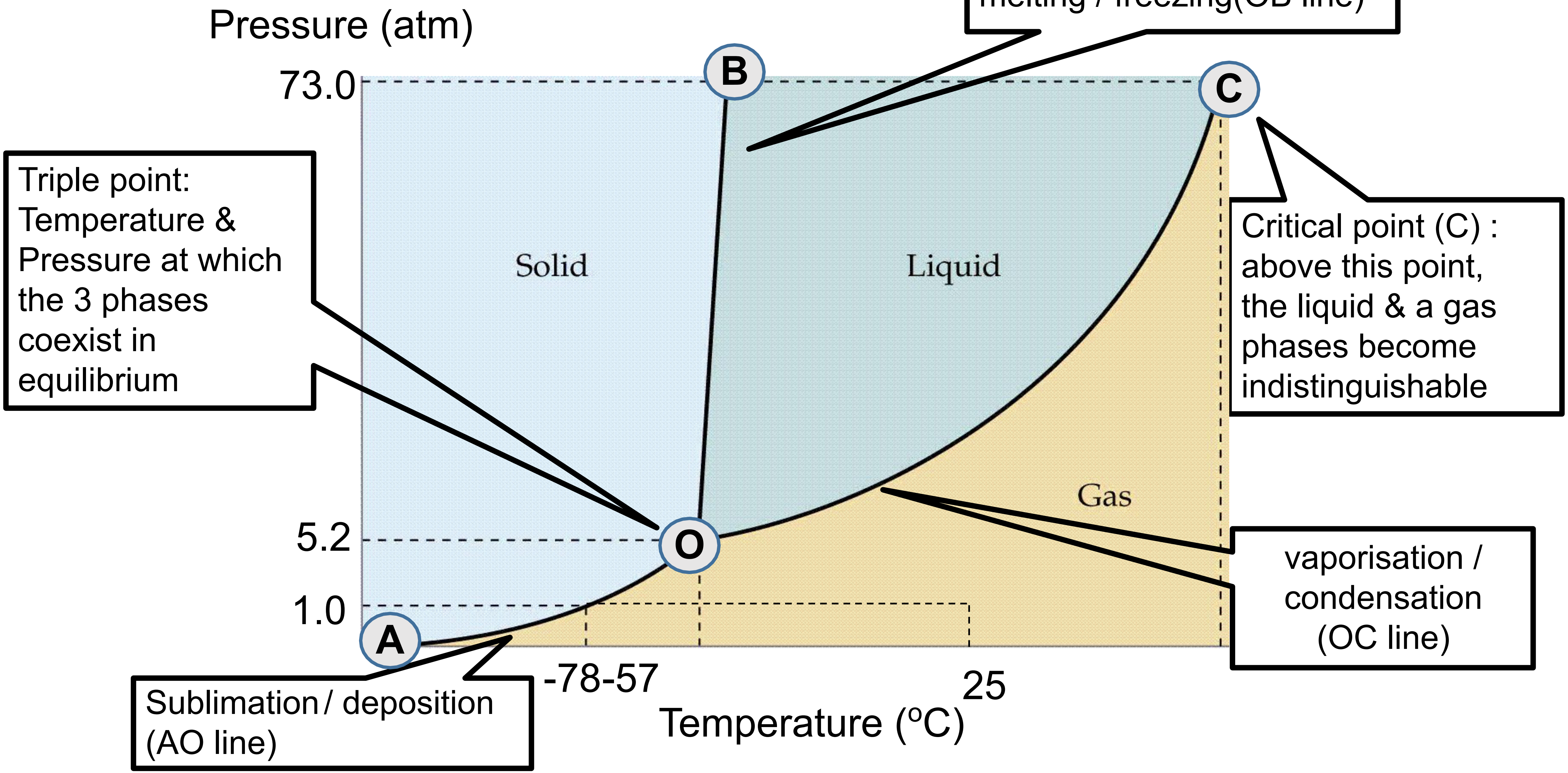
Triple Point

- Triple point is the point at which the three phases, solid, liquid and gas, can coexist at the same temperature and pressure.
- This is where three lines meet together

Critical Point

- Critical point is a point on the liquid-gas equilibrium curve, above which it is impossible to condense a gas into a liquid.
- The Liquid-gas line ends

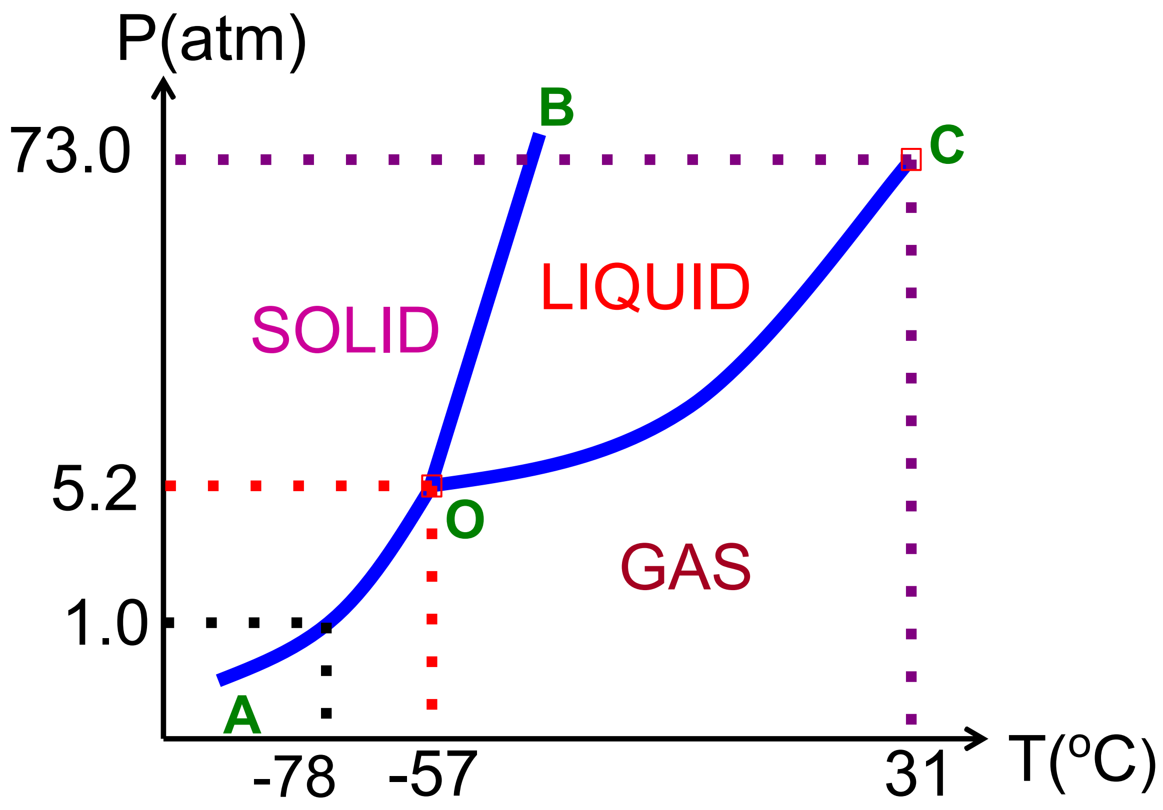
PHASE DIAGRAM OF CARBON DIOXIDE



CO_2 is a gas under normal conditions (25°C, 1 atm)

PHASE DIAGRAM OF CARBON DIOXIDE

Phase Diagram of CO_2 is a Typical Phase Diagram

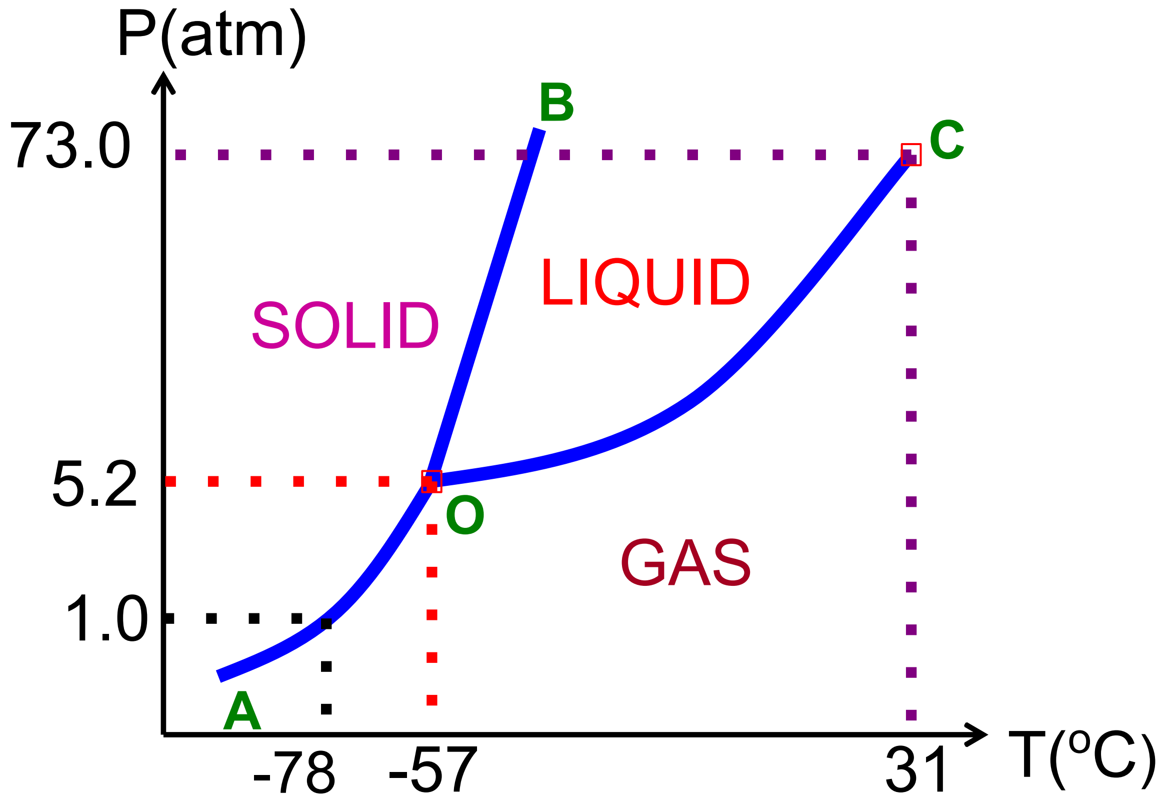


Explanation:

- At 1 atm, CO_2 is a solid at temperature below -78°C.
- Above this temperature, CO_2 is a gas.
- The triple point of CO_2 occur at -57°C and 5.2 atm.
- Below this point, CO_2 cannot exist as a liquid.
- at 1 atm solid CO_2 does not melt but sublime when heated. Because the normal pressure is below the triple point (5.2 atm), solid CO_2 sublimes without turning into liquid.

PHASE DIAGRAM OF CARBON DIOXIDE

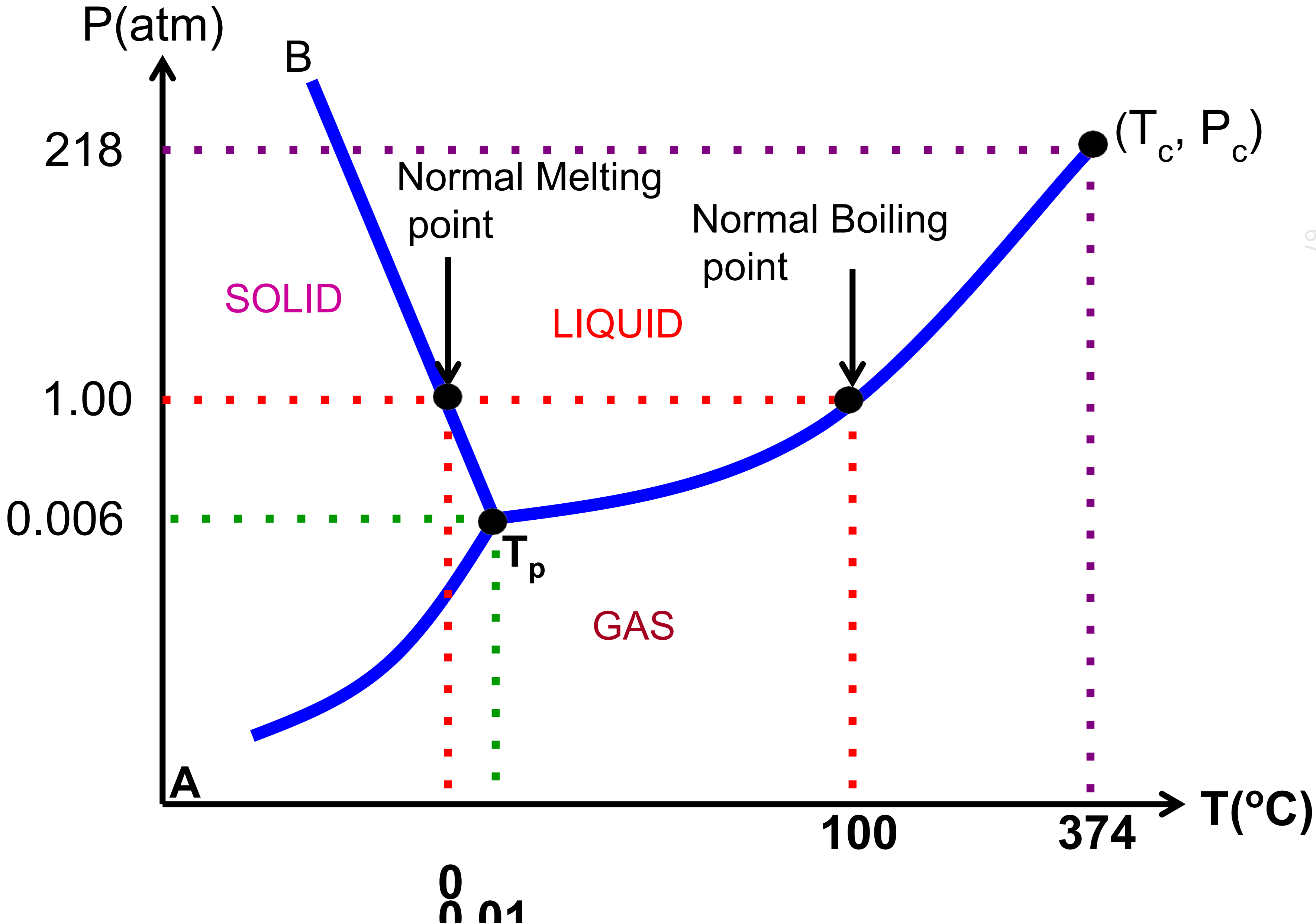
Phase Diagram of CO_2 is a Typical Phase Diagram



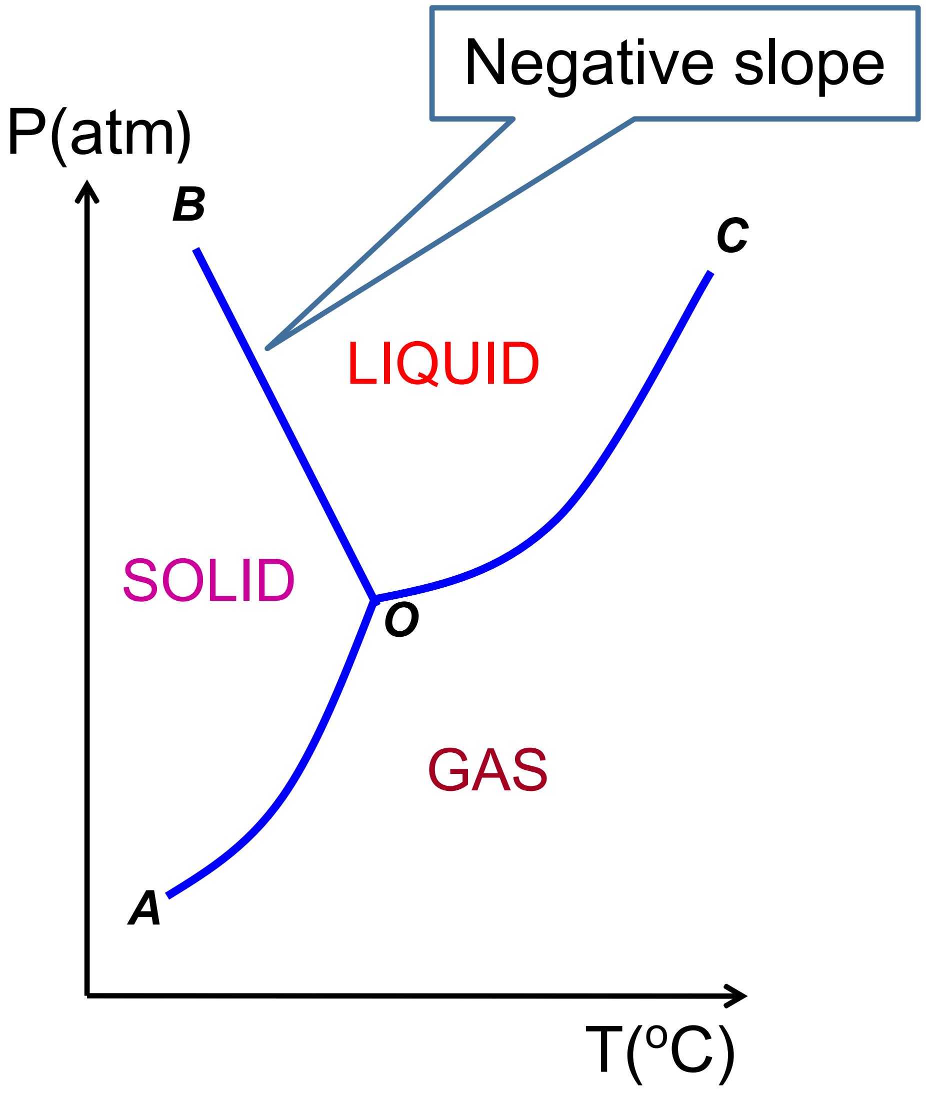
Explanation:

- Positive slope of solid-liquid equilibrium line means melting point increases when pressure is increased.
- When pressure is increased, formation of more compact solid CO_2 is favoured as solid CO_2 is denser than liquid CO_2.
- More difficult to melt the solid CO_2 at high pressure.

PHASE DIAGRAM OF WATER



PHASE DIAGRAM OF WATER



Anomalous Behaviour of Water

- Negative slope of OB lines means melting point decreases when pressure is increased.
- Ice has lower density than liquid water. It occupies bigger volume than liquid water

$$H_2O(s) \rightleftharpoons H_2O(s) \quad V_{ice} > V_{water}$$

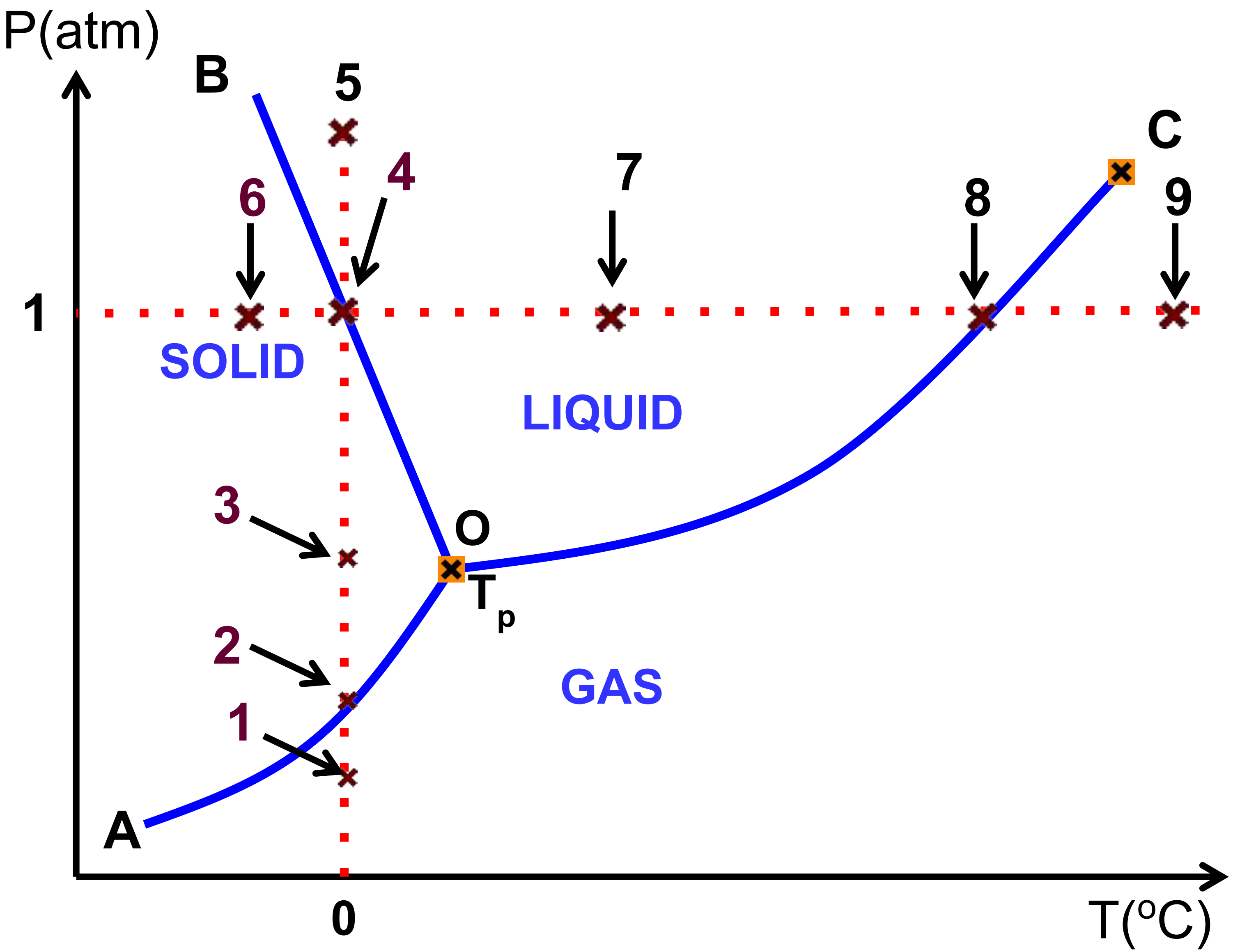
- When pressure is increased, formation of more compact liquid water is favoured
- Therefore, ice easily melts at high pressure.

EXAMPLE:

By referring to the phase diagram of water given, describe any changes in the phases present when H_2O is

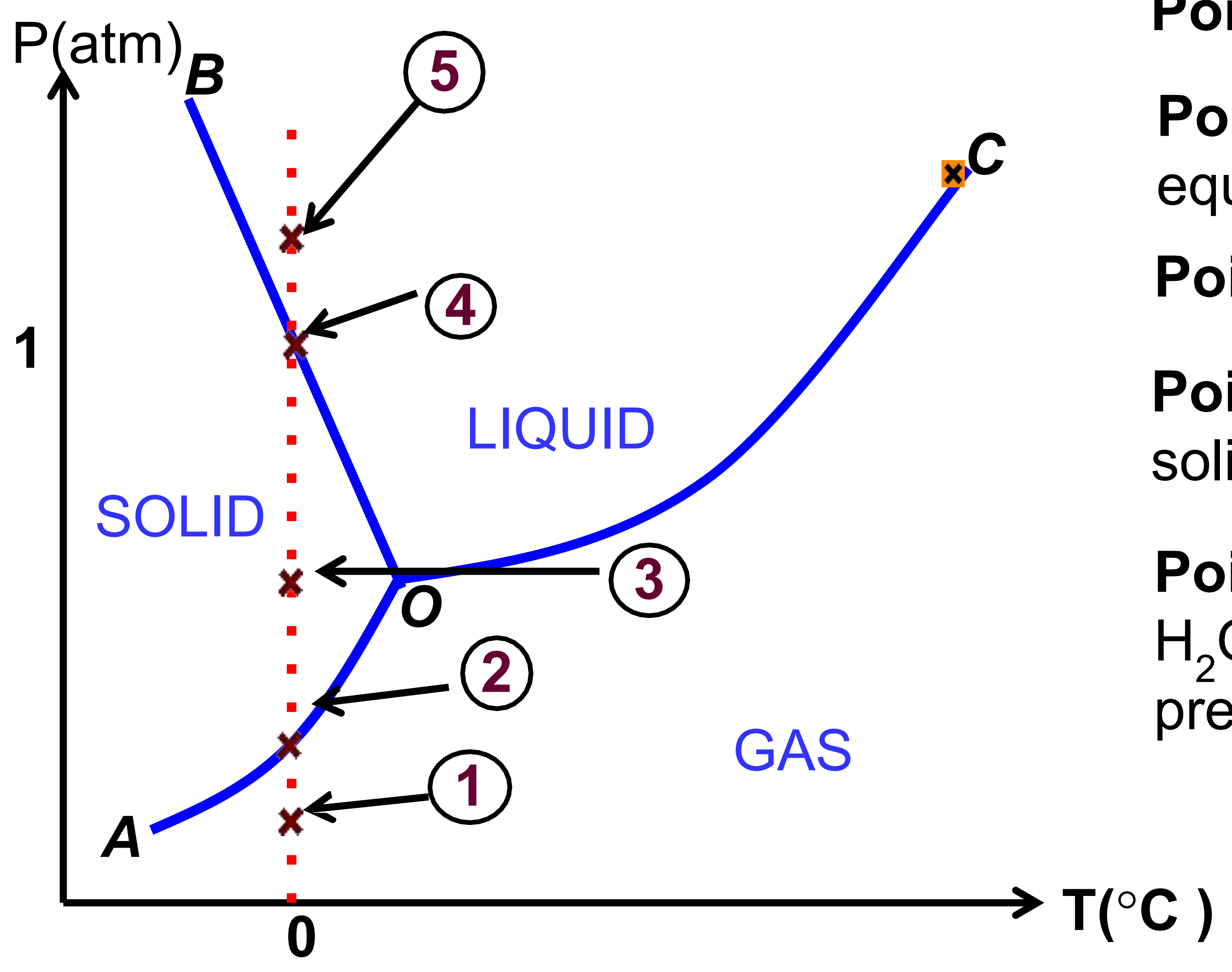
a) kept at 0°C while the pressure increased from point 1 to point 5 (vertical line)

b) kept at 1.00 atm while the temperature is increased from point 6 to point 9 (horizontal line)



SOLUTION:

a)



Point 1 : H_2O exists totally as a vapour

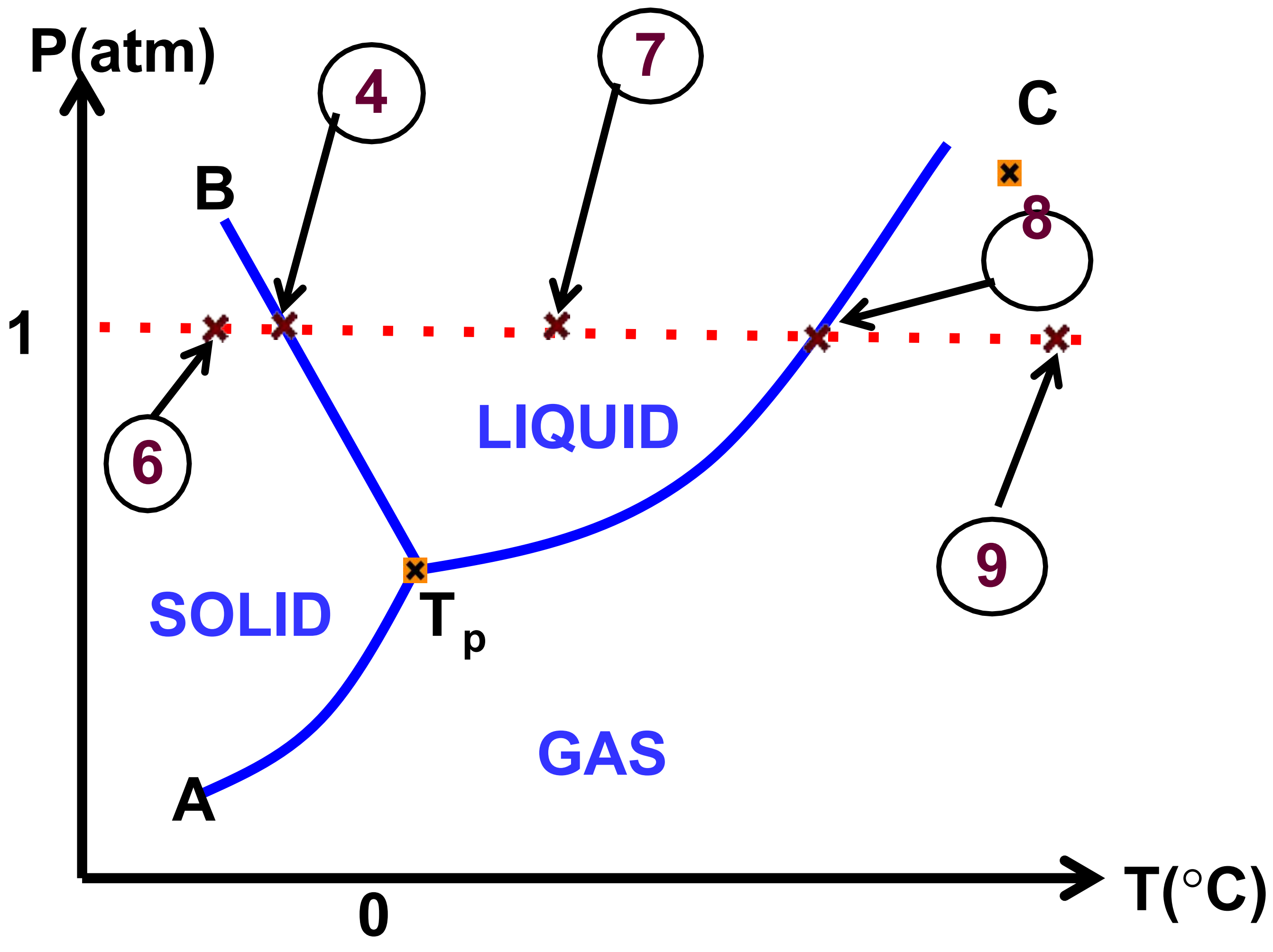
Point 2 : a solid-vapour equilibrium exists.

Point 3 : all H_2O is converted to solid

Point 4 : equilibrium between solid and liquid is achieved

Point 5 : As pressure increases, all H_2O melts. So, only the liquid phase is present

SOLUTION:



b) **Point 6** : H_2O exists entirely as solid

Point 4 : the solid begins to melt & equilibrium exists between the solid & liquid phases

Point 7 : At even higher temperature, the solid has been converted entirely to a liquid

Point 8 : a liquid-vapour equilibrium exists

Point 9 : upon further heating to point 9, H_2O is converted entirely to the vapour phase

THE END